

Convective Precipitation Nowcasting Using Deep Learning Integration of Satellite and Radar Imagery

submitted by

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Abstract

Convective precipitation is highly localized and evolves continuously over short timescales, making short-term precipitation forecasting particularly challenging despite its importance for early warning and weather-sensitive operations. This thesis evaluates five deep learning architectures for convective precipitation nowcasting over Germany and investigates whether combining radar and satellite observations improves forecasting performance compared with a radar-only configuration. The models were trained using paired RADOLAN radar and SEVIRI satellite observations collected between 2015 and 2024. In addition to the five individual architectures, an ensemble combining their predictions was also evaluated. Forecast performance was assessed at lead times of 15, 30, 45, and 60 minutes using both continuous and categorical evaluation metrics. No single architecture performed best across all metrics. VPTR achieved the strongest individual performance overall, particularly at longer lead times, while the ensemble was competitive at shorter horizons but its relative ranking declined as lead time increased, and it did not consistently outperform the best individual model. Integrating satellite observations generally improved forecast quality for the deep learning models, with the largest gains observed at longer prediction horizons, although this benefit did not extend to the extrapolation-based baseline method. Severe convective precipitation remained the most difficult to predict because such events were underrepresented in the training data. These findings demonstrate the importance of both model architecture and multimodal observations for improving convective precipitation nowcasting. The code and implementation used in this thesis are available on [GitHub](#).

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This thesis was completed during a period when I was also dealing with persistent insomnia. The many sleepless nights made the process more difficult, but they also taught me patience, perseverance, and resilience. Despite these challenges, I remained determined to finish my work. I am thankful to everyone who supported me, believed in me, and helped me continue until the end.

Berlin, September 12, 2026

Atikul Islam Sajib

Declaration

I hereby declare that I have written this thesis independently and without illegitimate assistance from third parties. Any parts adopted verbatim or in substance from other works have been identified as such, with the sources indicated. AI-based tools were used where applicable in accordance with the guidelines of the institution, and their use is documented in the AI Usage Statement included in this thesis.

Date

Signature

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Chapter 1

Introduction

Convective precipitation nowcasting predicts intense precipitation over the next one to two hours and has received increasing attention in recent years [1]. Convective events, such as thunderstorms, are associated with several weather conditions, including intense precipitation and hail. Reliable short term forecasts are therefore important for providing timely information before severe conditions develop. For practical use, these forecasts need to be produced at high spatial resolution and with minimal processing delay.

Meeting these requirements remains difficult, particularly when weather conditions change within a short period. Traditional forecasting methods have been extensively used for precipitation prediction [2, 3], but the increasing amount of weather observations has created new possibilities for data-driven forecasting. Instead of defining every forecasting relationship in advance, data-driven methods learn from previous observations, allowing future conditions to be estimated based on the identified patterns.

Deep learning has extended this direction and allows the direct processing of large sequences of weather observations [4]. Over time, different network designs have been introduced for precipitation nowcasting. Recurrent models were among the earlier approaches used for sequential prediction, followed by convolutional encoder-decoder networks [5] and, more recently, Transformer-based models [6]. Each model family handles the forecasting task differently, reflecting the continued development of learning based methods in this field.

At the same time, advances in weather observation systems have increased the amount and variety of data available for nowcasting. Radar and geostationary satellite instruments provide frequent observations that can be used to follow changing weather conditions [7]. The availability of these long term observational records has supported the training and evaluation of increasingly complex prediction models and has contributed to the continued development of precipitation nowcasting. The following sections describe the motivation behind this work, the specific contributions made, and the overall structure of the thesis.

1.1 Motivation

This work is motivated by the expanding number of deep learning models for precipitation nowcasting and the challenges in comparing them directly. Most models are evaluated on different datasets, training settings, and evaluation criteria, making it hard to determine whether improvements result from the model architecture or the experimental setup. A common evaluation setting is needed to better understand how selected architectural designs perform for convective precipitation nowcasting.

The increasing availability of geostationary satellite imagery has encouraged the development of multimodal approaches for convective precipitation nowcasting. As more studies have incorporated satellite observations alongside radar data, interest has developed in understanding how this additional information can improve predictions of convective precipitation. However, it remains unclear how much this improvement varies in magnitude across different deep learning architectures. This uncertainty is examined directly in this thesis by comparing radar-only and multimodal input across the same set of architectures.

1.2 Research Hypotheses

Based on the motivation outlined above, the following hypotheses are investigated in this thesis:

- H1:** The choice of deep learning architecture affects precipitation nowcasting performance under comparable experimental conditions.
- H2:** Combining radar and satellite observations improves precipitation nowcasting performance compared with using radar observations alone.
- H3:** An ensemble of multiple deep learning models improves precipitation nowcasting performance compared with individual architectures.

1.3 Contributions

The key contributions developed through this work are outlined below:

1. **Multimodal nowcasting dataset.** We assemble and preprocess a decade-long archive of paired RADOLAN radar and SEVIRI satellite observations over Germany (2015–2024) for convective precipitation nowcasting. The preprocessing pipeline includes spatial alignment, log-normalization, seasonal filtering, and quantile-based sampling to address extreme class imbalance.
2. **Systematic multi-architecture comparison.** We train and evaluate five deep learning architectures spanning recurrent (ConvLSTM), convolutional (SmaAt-UNet, SimVP),

and Transformer-based (Earthformer, VPTR) designs under a consistent experimental setup to compare the effect of architectural variations.

3. **Radar-satellite fusion across architectures.** We apply a consistent multimodal input strategy that fuses radar precipitation fields with two SEVIRI thermal infrared channels via channel concatenation across all five architectures. This enables a systematic assessment of whether satellite data improves forecasting performance across different model families.
4. **Loss function combining multiple components for convective nowcasting.** We combine a five-tier precipitation-weighted Mean Absolute Error (MAE), a gradient sharpness penalty, and horizon-dependent weighting into a single loss function, designed to address rainfall intensity imbalances and encourage spatially sharp forecasts of intense convective events.
5. **Ensemble evaluation.** We implement and evaluate a simple averaging ensemble of all five architectures to directly analyze whether combining predictions from multiple models improves forecast quality compared with individual architectures.

1.4 Thesis Structure

This thesis is structured into six chapters, organized as follows:

Chapter 2 reviews the relevant literature on precipitation nowcasting, covering traditional physics-based and extrapolation approaches, deep learning methods for radar and satellite nowcasting, and existing multimodal fusion strategies.

Chapter 3 details the datasets used in this study, describing the RADOLAN radar product and the SEVIRI satellite instrument alongside the preprocessing pipeline designed to ensure data consistency, normalization, and alignment.

Chapter 4 presents the methodology, formulating the sequence-to-sequence forecasting task, the input fusion strategy, the five selected deep learning architectures, the multi-component loss function, and the evaluation framework.

Chapter 5 reports the quantitative results of the experiments, presenting the performance of each model across all four forecast horizons using our evaluation metrics. Furthermore, it provides an in-depth discussion of the results, analyzing the operational strengths and structural limitations of each architectural family under varying convective conditions.

Chapter 6 concludes the thesis by summarizing the main findings, addressing the limitations of the current approach, and proposing directions for future research. It also reflects on the practical lessons learned throughout the project, which may be useful for similar future studies.

1.5 AI Usage Statement

During the preparation of this thesis, AI-based tools were used for specific and clearly defined supporting tasks. The following describes how each tool was used.

1. **Grammarly** was used for grammar checking, spelling correction, and sentence-level language refinement. The Grammarly Citation Finder feature was also used to locate potentially relevant references during the literature review.
2. **QuillBot** was used as a secondary rephrasing tool when a Grammarly-corrected sentence still did not read clearly or naturally. It was used selectively to find alternative formulations for specific sentences, and all rephrased output was reviewed and adapted by the author before being used.
3. **ChatGPT** (OpenAI) and **Claude** (Anthropic) were used to assist with debugging, correcting, refining, and refactoring Python code. Both tools also provided support for running experiments on the GPU cluster, including preparing and troubleshooting SLURM job scripts and resolving software-environment and dependency-related issues. For LaTeX, the tools were used to correct syntax and compilation errors, improve the formatting and alignment of equations, check the consistency of mathematical notation, and format tables, figures, labels, and cross-references. They were additionally consulted for suggestions concerning the structure, organisation, and ordering of thesis sections.
4. **NotebookLM** (Google) was used to assist with understanding research papers by extracting and summarising key ideas, including methodology, dataset construction approaches, and input-output design choices. This supported the literature review process and helped the author efficiently engage with a large number of papers. All content derived from this tool was independently verified by the author against the original papers and discussed with the supervisors before being used in the thesis.

All code used in this thesis was written by the author. AI tools were used only for the specific purposes described above, and were not used to generate thesis content or code implementations. The writing of all thesis chapters, the methodology, experimental design, data analysis, and interpretation of results were carried out by the author. All results, findings, methodology decisions, and next steps were discussed with and reviewed by the supervisors throughout the thesis.

Chapter 2

Fundamentals and Related Work

Precipitation nowcasting has become much more advanced in recent decades owing to advances in atmospheric observation and computing. Early precipitation nowcasting techniques relied on either numerical weather prediction (NWP) models or image extrapolation from existing radar observations [2, 3]. The limitations of early nowcasting techniques were due to the amount of data and the computing power available at the time. With advancements in radar technology and the availability of satellite data, it became possible to use machine learning algorithms, including deep learning methods, for precipitation prediction [8]. These models are capable of recognizing complex patterns within the data. Currently, radars and satellites are the most common data sources for nowcasting precipitation because these sources provide frequent, high resolution observations. Every nowcasting technique, from traditional methods to modern deep learning models, has its own advantages and disadvantages, making it useful for certain types of nowcasting. These approaches are discussed below (Sections 2.1, 2.2, 2.3, 2.4, and 2.5).

2.1 Challenges in Predicting Convective Precipitation

Convective precipitation is difficult to predict because storms can develop and change within minutes. Local atmospheric conditions, particularly humidity, strongly influence this development [9]; even small changes can affect where and when a storm forms and how intense it becomes. In addition, convective cells significant grow, weaken, split, or merge within a short period, adding further variability that is hard to anticipate.

Operationally, nowcasts must be produced quickly enough to support warnings and decision-making, which leaves only a limited window to process recent observations and generate predictions. This becomes harder at longer lead times, since storms continue to evolve after the forecast is issued, increasing uncertainty the further ahead the prediction reaches.

Detailed high resolution observations are therefore essential for capturing these rapid changes in convective precipitation [8]. Low-resolution data can miss small cells, sharp storm boundaries, and local variations in rainfall intensity, whereas high-resolution radar captures these features more clearly, which matters most for localized, heavy precipitation events where intensity can vary sharply over short distances.

2.2 Traditional Approaches for Precipitation Nowcasting

Before the application of machine-learning and deep-learning techniques to predict convective precipitation, short-term prediction was made using two classical approaches. The first approach is physics-based modeling, in which future weather conditions are predicted by simulating changes in variables such as motion, humidity, and pressure (Section 2.2.1). The second method is extrapolative forecasting, which assumes that the current rainfall pattern will continue into the future at the same rate (Section 2.2.2). These approaches existed long before the computational capabilities required for deep learning became available and are still commonly used as benchmarks for evaluating modern forecasting models.

2.2.1 Numerical Weather Prediction (NWP)-Based Methods

Numerical Weather Prediction (NWP) is the standard approach for weather forecasting and forms the basis of most operational forecasting systems [2]. It predicts future atmospheric conditions by solving the governing physical equations of the atmosphere using observations as the initial state. Although NWP provides accurate forecasts over many applications, predicting convective storms at very short lead times remains particularly challenging for several reasons.

The limitations of numerical weather prediction at very short lead times are largely related to the representation of the atmosphere's initial state. Even a minimal error in initializing a weather model can grow substantially over time making precise forecasts increasingly difficult regardless of the model's accuracy [10]. Obtaining an accurate initial state is particularly difficult for rapidly evolving convective systems, where small-scale changes can strongly affect the subsequent development of precipitation. When new observations are incorporated, the model takes time to adjust to a physically consistent atmospheric state. This adjustment can occupy a substantial portion of the nowcasting window, limiting the time available for useful prediction.

To improve the model's initial conditions, several methods have been developed. These methods incorporate real observations directly into storm-scale models before a forecast begins. While these approaches can improve the resulting forecasts, their benefits remain constrained by the complexity of the atmospheric processes and the spatial resolution of the modeled domain. For example, a large field study found that local convective storms were particularly difficult to predict, with the predictability of precipitation varying substantially depending on the prevailing meteorological conditions and storm characteristics [11].

Computational demand is another important practical limitation of NWP. Representing individual storms requires a fine spatial grid, and running the model at this resolution is computationally expensive, making frequent updates difficult. This constraint can also limit the number of ensemble members and the frequency at which ensemble forecasts can be generated, reducing the ability to represent forecast uncertainty in operational settings. For this reason, NWP forecasts often cannot be produced fast enough to meet the operational demands of real-time nowcasting.

2.2.2 Extrapolation Methods

Instead of modeling atmospheric physics, extrapolation methods use the most recent radar images to extend the existing rain pattern forward in time. The assumption is that precipitation does not change substantially over short periods, making its recent evolution a reliable basis for short term prediction. If its speed and direction are known, future radar images can be estimated by shifting the current image along the estimated path.

One of the best known extrapolation approaches is Lagrangian extrapolation [3]. In this method, the motion of rainfall is estimated from a series of past radar images. The latest image is then shifted along the estimated motion field to produce a forecast. This method became a fundamental reference for many advances in rainfall prediction and remains the foundation for several modern nowcasting systems. Building on this, open source pySTEPS [12] toolkits have made these techniques more accessible by combining several motion tracking methods, including optical flow algorithms, with stochastic components that introduce randomness to better represent the uncertainty of storm evolution. These optical flow approaches were also evaluated against the operational nowcasting system of the German Weather Service and achieved comparable or better performance in several cases [13].

Despite these improvements, the skill of extrapolation methods is generally limited to lead times of about 30 minutes. Moreover, these methods are computationally efficient, where the assumption of continuous precipitation motion is reasonable. However, extrapolation cannot capture the formation of new storms, the dissipation of existing ones, or sudden changes in rainfall intensity, thereby limiting forecast accuracy for convective precipitation.

2.3 Radar-Based Precipitation Nowcasting

Radar weather tracking uses radio wave emissions into the air and records the reflected energy from raindrops. The reflected energy determines rainfall intensity and helps measure the actual rainfall rate. The major advantage of radar technology over conventional rain gauges is the coverage. While conventional rain gauges track rainfall at only one point, radar covers enormous areas at once. It allows meteorologists to have a picture of a storm and its development in real time. Since radar produces highly detailed pictures updated every several minutes, it is used extensively to feed machine learning and deep learning algorithms (Sections 2.3.1 and 2.3.2), which help track the storm's movement and predict rainfall a few hours ahead.

2.3.1 Machine Learning Approaches

Before deep learning gained popularity, traditional machine learning methods were widely used for short term precipitation prediction. Unlike conventional forecasting approaches, these methods learned the relationship between meteorological observations and precipitation directly from historical data, allowing them to capture complex patterns with-

out explicitly modeling the underlying physical processes. This data driven framework formed the basis of many early studies and demonstrated the potential of machine learning for improving precipitation forecasting.

Several precipitation forecast related applications have used machine learning techniques to solve various problems such as convective rainfall forecasting [14], and precipitation return period estimation [15]. In order to solve these problems, different classification and regression methods have been applied depending on the specific dataset and forecasting requirements. The results of these studies indicate that machine-learning methods can provide accurate forecasts for various precipitation prediction tasks when suitable input data are available. In some cases, they have also performed better than the traditional approaches (Section 2.2).

Despite these successes, traditional machine learning methods share a common limitation, for example, these approaches depend on handcrafted features such as past precipitation amounts, geographic locations, and brightness temperatures. While computationally inexpensive and interpretable, they generally perform best at short lead times, struggle to represent complex spatial patterns of convective storm formation, and tend to degrade when applied to large, high-dimensional datasets.

2.3.2 Deep Learning Approaches

Deep learning techniques have become increasingly common in precipitation nowcasting compared with traditional machine learning methods due to the availability of large radar datasets. Unlike traditional machine learning algorithms, these methods learn the spatial and temporal patterns from radar images. This feature makes them better at representing precipitation systems and their dynamics. Therefore, deep learning techniques have been adopted for a variety of precipitation nowcasting tasks, including rainfall occurrence prediction as well as forecasting the movement and intensity of precipitation systems.

Early deep learning methods for precipitation nowcasting focused on learning the spatiotemporal evolution of radar observations. One of the first approaches was ConvLSTM [16], which incorporated convolutional operations into an LSTM [17] framework to combine model spatial and temporal information. Compared with fully connected LSTM models and the optical flow methods, ConvLSTM produced more accurate precipitation forecasts as this architecture can capture long term patterns.

Encoder-decoder architectures based on U-Net [18] have been widely adopted later for radar-based precipitation forecasting because these architectures effectively preserve both local and global spatial features through their hierarchical design and skip connections. RainNet is one representative model that applies this architecture to precipitation nowcasting using quality controlled radar data from the German Weather Service [19]. As research on U-Net-based nowcasting continued, several studies focused on improving different aspects of the architecture. Among them, SmaAt-UNet emphasized computational efficiency by incorporating attention modules and depthwise separable convolutions, achieving forecasting performance comparable to considerably larger networks while requiring substantially fewer parameters [5].

Generative models have been used in precipitation nowcasting to address challenges of

deterministic forecasting, such as producing overly smooth forecasts and failing to capture precipitation uncertainty. Generative adversarial networks (GANs) have been used to generate more realistic precipitation fields while maintaining strong forecasting performance [20]. Later, the focus shifted toward generative models that better capture the evolution of precipitation systems over time. For example, NowcastNet combines a physics-based motion model with a generative network to improve predictions of localized, rapidly evolving heavy rainfall [21]. More recently, diffusion models have explored precipitation nowcasting to improve forecast quality and provide a better representation of forecast uncertainty [22].

While deep learning techniques have improved performance, there are still a number of limitations. Most models have difficulty forecasting intense or localized rainfall events and tend to produce smooth forecasts that fail to preserve the structure of precipitation. Furthermore, forecasting accuracy decreases with increasing lead time and increasing blur in the predicted image.

2.4 Satellite-Based Precipitation Nowcasting

Weather satellites detect two sources of energy such as sunlight reflected from clouds and heat radiated by the Earth and its atmosphere, which form the basis for cloud observation. Geostationary satellites capture new observations at short time intervals, allowing changes in weather systems to be monitored over time. While ground based radar is limited to a fixed coverage area, satellites can observe oceans, mountains, and other regions where radar data is unavailable, providing a much wider view of how weather systems form and move. Therefore, the wide coverage and frequent availability of satellite data make it useful for precipitation and storm forecasting, as demonstrated by the machine learning and deep learning studies (Sections 2.4.1 and 2.4.2).

2.4.1 Machine Learning Approaches

Radar-based approaches have long dominated nowcasting research; machine learning techniques have also been extended to satellite data to address radar's coverage limitations. Unlike radar, satellites do not measure rainfall directly. Therefore, earlier approaches first extracted cloud properties, such as cloud-top temperature, thickness, and water content, before using them as inputs to machine learning models. Therefore, these features provided the information needed to estimate precipitation from satellite observations and predict precipitation.

Machine learning approaches have been implemented to forecast precipitation using satellites alone or combined with radar and lightning data. These include rainfall estimation [23], precipitation classification [24], and thunderstorm nowcasting [25]. In these cases, the relationship between cloud variables and precipitation is highly complex and non linear, making it difficult to model using predefined physical or statistical approaches. Hence, classical machine-learning methods were applied to satellite observations to identify complex relationships between the input variables and the target outcomes across different variables. Their ability to learn nonlinear patterns from the

data made them suitable for this application.

Despite these improvements, a fundamental limitation remains, similar to that of radar (Section 2.3.1). Since traditional machine learning models depend on hand-crafted features, their performance is bounded by the quality and representativeness of those features. Complex spatial and temporal patterns in the data can be difficult to capture, and performance tends to degrade under dynamic or unseen meteorological conditions. Moreover, the feature engineering process itself also requires substantial domain expertise and may miss interdependencies that are difficult to describe explicitly.

2.4.2 Deep Learning Approaches

High-resolution satellite data has made deep learning increasingly useful for short-term convective precipitation nowcasting. Satellite observations can provide information about developing weather systems before precipitation becomes fully established, making them useful for identifying and forecasting emerging convective events. Deep learning approaches have therefore been applied to satellite observations to improve precipitation forecasting, with a particular focus on the evolution of cloud systems across successive observations.

In satellite based forecasting, consecutive images provide information about the development and movement of cloud systems. Recurrent neural networks are well suited to processing such sequential observations because these methods retain information from earlier inputs when analyzing subsequent images. Several recurrent architectures have therefore been explored for satellite based precipitation forecasting. Among these, PredRNN has been applied to satellite image sequences for rainfall forecasting [26]. Nevertheless, preserving detailed precipitation structures remains challenging, particularly during significantly developing rainfall events. To address this problem, TUPANN models precipitation motion and intensity separately, allowing rainfall structures to be represented more clearly and improving forecasting accuracy [27].

Large scale precipitation forecasting requires capturing information over wider areas and extended time periods, creating challenges for deep learning models. Transformer-based architectures were developed to capture these dependencies. For example, SaTformer applies Transformers to satellite image sequences for effective precipitation nowcasting [28], while its satellite based approach enables prediction over much larger areas than ground based radar. To further improve large scale operational forecasting, researchers developed hybrid deep learning models that combine the strengths of different neural network architectures. These models integrate various meteorological data into a single forecasting model. The Global MetNet uses a hybrid approach, combining satellite data and global weather data to produce high resolution global precipitation forecasts [29].

Despite these advances, some important challenges remain. Satellite data is not available in all regions, limiting the applicability of satellite based forecasting. The development, operation, and maintenance of satellite systems also require substantial financial and technical resources, which can limit the availability and continuity of satellite observations. In addition, satellite based models can produce smooth or blurry precipitation predictions because the relationship between cloud properties and surface rainfall is complex, as similar cloud properties can produce different amounts and patterns of rainfall.

As a result, these models may have difficulty capturing small-scale changes and sharp precipitation boundaries in localized storms.

2.5 Multimodal Deep Learning Approaches

Although many precipitation forecasting models still depend on a single input source, recent studies have increasingly explored the combined use of multiple observations. This approach, known as multimodal learning, allows a model to use complementary information from different data sources within a single forecasting framework. However, combining heterogeneous observations effectively remains an important challenge in model design. The following subsections discuss general techniques for multimodal data fusion (Section 2.5.1) and focus specifically on the integration of radar and satellite observations (Section 2.5.2).

2.5.1 Data Fusion Techniques

When multiple input sources are available, effectively combining their information is an important part of model design. The stage at which the information is combined influences how the model learns from different inputs and how well complementary information can be utilized. A suitable multimodal fusion strategy enables the model to make better use of information from each source, leading to more reliable predictions.

The simplest approach is to combine all inputs at the start, allowing the neural network to learn combine feature representations from a unified input. This enables the model to integrate information from all input sources from the earliest stages of processing. However, the model must figure out how much weight to provide each source, which becomes harder when inputs differ significantly in resolution and physical meaning. As a result, the shared representation often fails to fully capture the complementary information available from different input sources [30].

In intermediate fusion, each input source is processed independently to extract modality specific features, which are then integrated during the later stages of the network. This allows the model to extract source specific information before integrating complementary features from different observations. This strategy generally produces more informative feature representations but at the cost of increased network complexity [31].

An alternative multimodal fusion strategy uses attention mechanisms to guide the fusion process. Instead of performing fusion at a predefined stage, the model adaptively selects the most informative features from each input source. This enables the attention mechanism to assign higher weights to informative regions while down-weighting less relevant or noisy information. Therefore, the result at prediction stage is a more effective integration of complementary data sources [32].

The effectiveness of a fusion method depends on the information provided by the input sources and the requirements of the forecasting task. A method that works well with one combination of inputs may not provide the same benefit when different sources are used. For this reason, different fusion strategies should be tested under the same conditions to

determine which method makes better use of the available information for precipitation nowcasting.

2.5.2 Radar–Satellite Data Integration

As discussed in (Sections 2.3 and 2.4), radar and satellite observations describe different aspects of precipitation systems. Because these two data sources provide complementary information, recent studies have explored combining them within deep learning-based multimodal forecasting frameworks.

Some of these multimodal models have focused on improving forecasting capabilities over longer lead times. Extending the forecast horizon is essential for capturing how convective precipitation develops and evolves, which increases the practical value of these models in real-world forecasting applications. The MetNet family is a representative example of this approach, integrating radar and satellite observations to produce precipitation forecasts. The original MetNet model extended lead times to 8 hours [33], and its successor, MetNet-2, extended this further to 12-hour forecasts [34]. The integration of radar and satellite observations has also been applied to other forecasting tasks, particularly to identify developing convective events such as thunderstorms at an early stage. For example, multimodal data fusion models combine radar, satellite, and additional meteorological observations to detect thunderstorms before clear precipitation signatures appear in radar observations [35].

Combining these two data sources is not straightforward in practice. Radar and satellite instruments differ in spatial resolution, update frequency, geographical coverage, and viewing geometry. These differences must be considered during preprocessing to ensure that observations from both sources represent the same location and time. The datasets must therefore be aligned onto a common spatial grid and matched according to their observation times before being used as inputs to a multimodal forecasting model.

2.5.3 Summary

Convective precipitation nowcasting has progressed from physics-based models and radar extrapolation to machine-learning and deep-learning methods. Although numerical weather prediction is effective over longer forecast horizons, it has limitations at very short lead times. Similarly, radar extrapolation performs well for brief periods but cannot adequately represent the formation, growth, and decay of convective cells. Traditional machine-learning methods improved data-driven forecasting; however, such methods remained dependent on manually selected features.

Deep learning reduced this dependence by learning spatial and temporal patterns directly from weather observations. Moreover, recent architectures have improved the representation of intense precipitation through attention mechanisms, advanced model designs, and generative approaches. Multimodal models have further extended this development by combining radar observations with satellite information about cloud conditions. Nevertheless, predicting localized heavy rainfall remains difficult. Direct comparison between studies is also limited because of differences in datasets, evaluation metrics, and experimental conditions.

Chapter 3

Dataset and Data Preparation

This study is based on a multimodal dataset comprising weather radar and geostationary satellite observations over Germany collected between 2015 and 2024. The two data sources provide complementary information that enhances the modeling of convective precipitation by combining radar and satellite observations within a multimodal model. The following sections describe the characteristics and acquisition of the radar and satellite datasets, examine their statistical properties to assess data quality and precipitation characteristics (Section 3.3), and present the preprocessing procedures required to align and prepare the data for model development (Section 3.4). In addition, the construction of the input and output sequences (Section 3.5), the temporal partitioning adopted for training, validation, and testing (Section 3.6.3), and the threshold-based methodology used to identify convective events and generate the target labels (Section 3.7) are described.

3.1 Radar Data

The radar data used in this study are obtained from the RADOLAN product, distributed through the DWD open data portal [36]. The dataset consists of quality-controlled radar-based precipitation composites covering Germany. It provides continuous observations at regular 5-minute intervals with a spatial resolution of $1\text{ km} \times 1\text{ km}$ and image dimensions of 1100×900 pixels.

To obtain quantitative precipitation measurements, the radar observations were converted to precipitation intensity in millimetres per hour (mm/h) and organized for efficient processing during model training. The high spatial and temporal resolution of the dataset enables the analysis of the evolution of precipitation systems and supports the learning of the spatial and temporal characteristics of convective precipitation. The main characteristics of the radar dataset are summarized in Table 3.1.

Table 3.1: Overview of the RADOLAN radar dataset

Description	Calculation	Result
Total days (2015–2024)	$365 \times 10 + 3 \text{ leap days}$	3,653 days
Images per day	$(24 \times 60) \div 5$	288 images
Total composites	$288 \times 3,653$	1,052,064 images

The radar composite count represents the maximum expected number of observations over the 2015–2024 period under the nominal acquisition schedule.

Figure 3.1 presents an example sequence of radar precipitation composites over Germany, illustrating the spatial distribution and temporal evolution of rainfall intensity across sequential time steps.

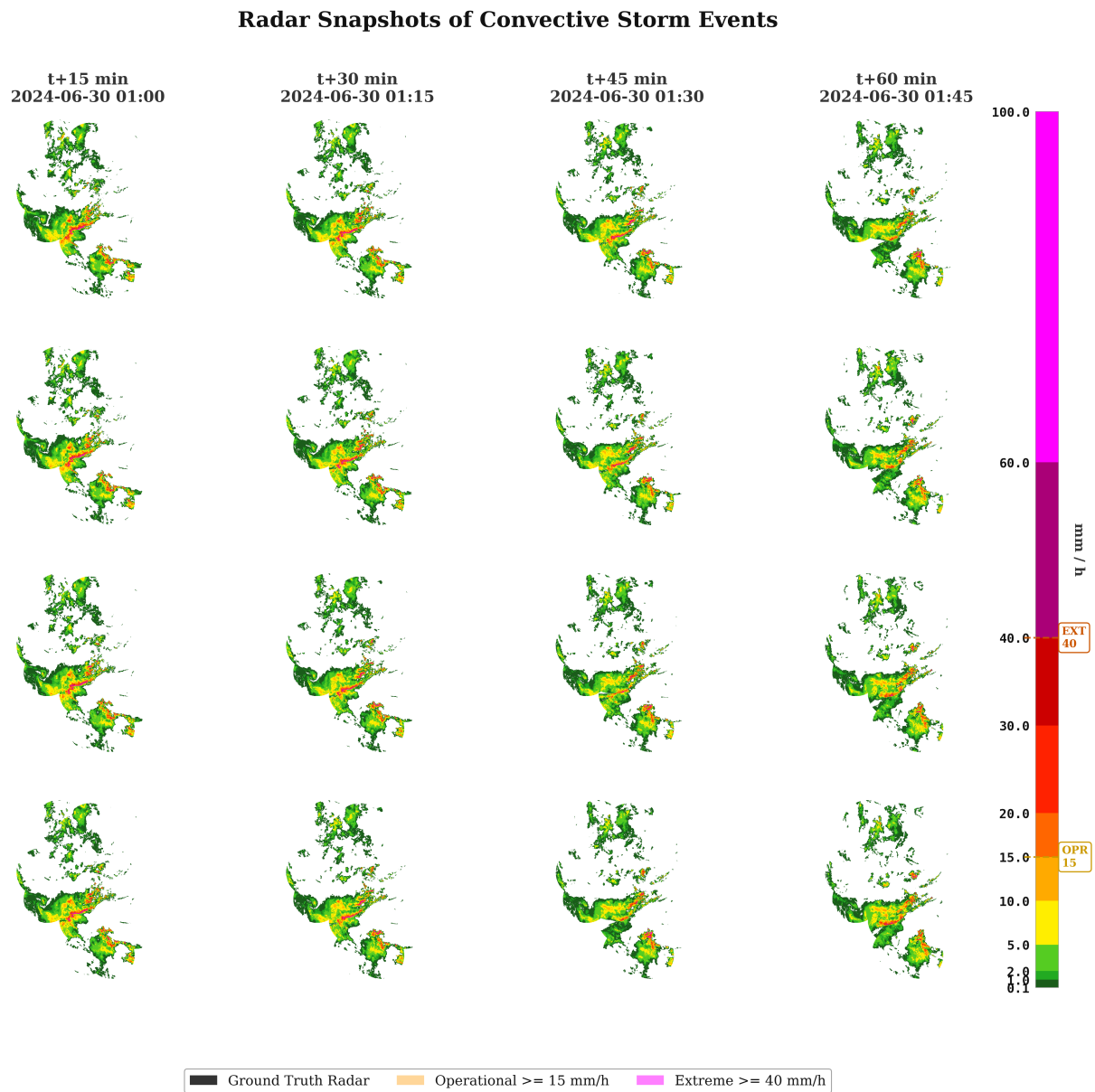


Figure 3.1: Radar precipitation composites over Germany during a convective storm event on 30 June 2024 (01:00–01:45 UTC).

Different precipitation intensity ranges are represented by distinct colors in the radar composites, allowing rainfall intensities to be distinguished during visual inspection. Dry areas are shown in ■ (< 0.1 mm/h), light rainfall appears in ■—■ (0.1–10 mm/h), and increasingly intense precipitation is represented by ■ to ■ (10 to > 60 mm/h). High-intensity precipitation is concentrated in localized convective cells, while the sequence of images illustrates the temporal evolution and movement of precipitation systems across the forecast horizons of 15, 30, 45, and 60 minutes.

3.2 Satellite Data

Satellite observations were acquired from the Meteosat Second Generation (MSG) High Rate SEVIRI instrument through the EUMETSAT data portal [37]. For consistency with the radar observations, the data were cropped to cover Germany and the surrounding region. SEVIRI provides observations in 12 spectral channels. Among these, Channels 7 and 9 were selected for this study because of their relevance to convective precipitation. Channel 7 (IR $8.7\mu\text{m}$) provides information on cloud properties, while Channel 9 (IR $10.8\mu\text{m}$) provides cloud-top temperature measurements that support the characterization of convective cloud systems.

Each satellite image has a spatial resolution of 175×320 pixels and is acquired at regular 5-minute intervals. The resulting sequence provides a consistent temporal record for each selected channel throughout the observation period. The number of SEVIRI satellite images available for each selected channel is summarized in Table 3.2.

Table 3.2: Overview of the SEVIRI satellite dataset

Description	Calculation	Result
Total days (2015–2024)	$365 \times 10 + 3$ leap days	3,653 days
Images per day per channel	$(24 \times 60) \div 5$	288 images
Selected channels	CH7 and CH9	2 channels
Total images per channel	$288 \times 3,653$	1,052,064 images
Total images (CH7 & CH9)	$1,052,064 \times 2$	2,104,128 images

Image totals were estimated based on a continuous 5-minute acquisition frequency for each channel over the entire 2015–2024 observation period, including leap years.

Figure 3.2 shows the selected infrared observations and their brightness temperature difference for four consecutive time steps. The sequence illustrates changes in the location and extent of cloud systems between successive 5-minute observations.

SEVIRI RSS - Germany (CH7, CH9, CH9 – CH7)

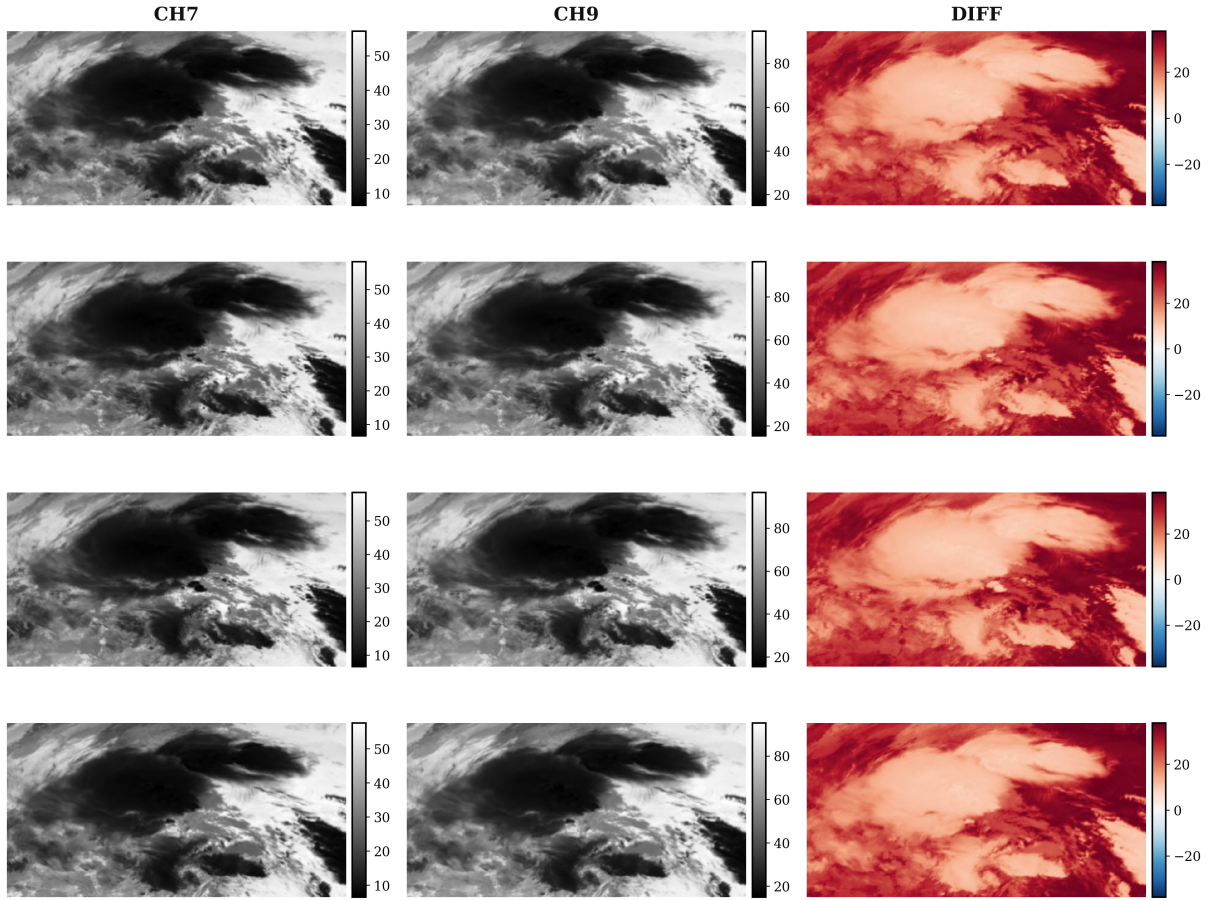


Figure 3.2: SEVIRI RSS observations over Germany illustrating changes in cloud patterns across four consecutive 5-minute observations.

Darker regions ■ in Channels 7 and 9 correspond to lower brightness temperature values, typically associated with cold cloud tops and deep convective clouds, whereas brighter regions correspond to higher values, representing warmer surfaces or shallow clouds. In the brightness temperature difference image (CH9–CH7), negative differences are shown in ■, while positive differences are shown in ■. These differences provide additional information on the thermal characteristics of the observed cloud systems.

3.3 Dataset Statistics

Before training the deep learning models, the radar and satellite datasets were statistically analysed to understand their quality and main characteristics. The analysis considered the spatial distribution and intensity of precipitation, together with changes in the data across different years and precipitation events. This helped to understand the data before the model training and to identify aspects that needed attention during data preparation. The results of these analyses are discussed in the following subsections.

3.3.1 Data Availability

Table 3.3 compares the expected and recorded numbers of radar and satellite observations for each year. A clear difference can be seen between the two datasets. The radar observations are almost complete over the entire study period, with only 0.1% missing in 2018. Unlike the radar dataset, the satellite dataset contains missing observations in every year. The highest percentages of missing observations occur in 2022 (22.5%), followed by 2017 (20.1%) and 2021 (18.8%). According to the EUMETSAT RSS catalogue, these missing observations were reported because of temporary interruptions in the Rapid Scan Service data acquisition [37].

Table 3.3: Availability of data on different radar and satellite datasets.

Year	Expected	Radar Avail.	Radar Miss.(%)	Sat. Avail.	Sat. Miss.(%)
2015	105,120	105,120	0.0%	92,664	11.8%
2016	105,408	105,408	0.0%	92,231	12.5%
2017	105,120	105,120	0.0%	83,986	20.1%
2018	105,120	105,060	0.1%	102,923	2.1%
2019	105,120	105,120	0.0%	104,710	0.4%
2020	105,408	105,408	0.0%	97,220	7.8%
2021	105,120	105,120	0.0%	85,386	18.8%
2022	105,120	105,120	0.0%	81,449	22.5%
2023	105,120	105,120	0.0%	90,865	13.6%
2024	105,408	105,384	0.0%	89,542	15.1%

Availability values represent the number of observations recorded in each dataset for the corresponding year. Missing values are given as percentages of expected observations.

3.3.2 Radar NaN Analysis

Around 39–43% NaN values were observed across all pixels each year. The minimum and maximum rates were 39.16% in 2015 and 43.34% in 2023. This shows the percentage of missing data is fairly constant year to year. The similar NaN rates across the years indicate that the missing values are not concentrated in a particular year. Therefore, an appropriate strategy for handling NaN values needs to be incorporated into the preprocessing pipeline before model development.

Table 3.4: Year-wise distribution of NaN values in the RADOLAN radar dataset.

Year	Total Pixels	NaN Pixels	NaN (%)
2015	104,068,800,000	40,750,907,928	39.16
2016	104,353,920,000	41,321,369,265	39.60
2017	104,068,800,000	40,944,601,010	39.34
2018	104,009,400,000	44,479,997,561	42.77
2019	104,068,800,000	44,826,856,010	43.07
2020	104,353,920,000	45,137,937,499	43.25
2021	104,068,800,000	44,986,338,827	43.23
2022	104,068,800,000	44,855,047,388	43.10
2023	104,068,800,000	45,099,765,241	43.34
2024	104,330,160,000	44,859,302,303	43.00

The NaN percentage is calculated as the ratio of NaN pixels to the total number of radar pixels for each year, computed over the full annual file set, consistent with Tables 3.5 and 3.6.

3.3.3 Rainfall Intensity Distribution

Table 3.5 shows the yearly distribution of rainfall intensity over Germany from 2015 to 2024. A similar pattern is observed in every year. Around 92% to 95% of all pixels record no rainfall, while most of the remaining pixels correspond to light rainfall. Moderate and heavy rainfall occur only in a small fraction of the observations. This distribution changes very little over the study period, indicating that heavy rainfall events are rare. The large difference between dry and rainy pixels, particularly between light and heavy rainfall, provides the basis for the sampling strategy and loss function described (Section 4.3.1).

Table 3.5: Annual RADOLAN Rainfall Intensity Distribution.

Year	No Rain (<0.1)		Light (≥ 0.1 – <2.5)		Moderate (≥ 2.5 – <10)		Heavy (≥ 10)	
	Pixels	%	Pixels	%	Pixels	%	Pixels	%
2015	59,570,006,153	94.1	3,215,680,411	5.1	498,018,981	0.8	34,186,527	0.1
2016	59,305,331,860	94.1	3,217,115,756	5.1	475,233,536	0.8	34,869,583	0.1
2017	58,928,985,980	93.4	3,564,145,950	5.6	589,597,821	0.9	41,469,239	0.1
2018	56,482,051,741	94.9	2,628,226,445	4.4	386,117,875	0.6	33,006,378	0.1
2019	55,267,538,243	93.3	3,465,585,752	5.8	476,522,649	0.8	32,297,346	0.1
2020	55,446,878,578	93.6	3,256,612,651	5.5	482,500,071	0.8	29,991,201	0.1
2021	55,061,321,174	93.2	3,479,538,376	5.9	502,400,978	0.9	39,200,645	0.1
2022	55,798,283,530	94.2	2,941,143,714	5.0	440,881,812	0.7	33,443,556	0.1
2023	54,383,048,118	92.2	3,958,981,047	6.7	587,552,063	1.0	39,453,531	0.1
2024	55,298,899,434	93.0	3,544,520,828	6.0	585,170,644	1.0	42,266,791	0.1
Total	565,542,344,811	93.6	33,271,550,930	5.5	5,023,996,430	0.8	360,184,797	0.1

Rainfall thresholds are expressed in millimetres per hour (mm/h). Pixel counts are aggregated over all 5-minute radar composites, computed over valid, non-NaN pixels only.

3.3.4 Precipitation Threshold Analysis

Table 3.6 presents the percentage of radar pixels that met or exceeded the selected rainfall thresholds for each year. Pixels with at least 0.1 mm/h indicate recordable rainfall, while values of at least 10.0 mm/h represent severe convective rainfall. Across all years, less than 0.1% of pixels reached or exceeded 10.0 mm/h, showing that extreme rainfall was infrequent. The highest percentages at the 0.1 and 1.0 mm/h thresholds occurred in 2023, while the highest percentages at 5.0 and 10.0 mm/h occurred in 2024. The lowest percentages at the first three thresholds occurred in 2018, while the lowest percentage at 10.0 mm/h occurred in 2020.

Table 3.6: Annual precipitation threshold analysis.

Year	≥ 0.1 mm/h (%)	≥ 1.0 mm/h (%)	≥ 5.0 mm/h (%)	≥ 10.0 mm/h (%)
2015	5.919	2.140	0.302	0.054
2016	5.913	2.058	0.296	0.055
2017	6.646	2.439	0.370	0.066
2018	5.119	1.657	0.264	0.055
2019	6.709	2.107	0.297	0.055
2020	6.365	2.090	0.301	0.051
2021	6.806	2.174	0.337	0.066
2022	5.768	1.889	0.293	0.056
2023	7.777	2.602	0.369	0.067
2024	7.015	2.480	0.376	0.071

Percentages represent the proportion of valid, non-NaN radar pixels exceeding each threshold across all 5-minute RADOLAN composites for the corresponding year.

3.3.5 Seasonal Distribution of Rainfall

Table 3.7 presents seasonal variation in mean rainfall intensity (mm/h) and total accumulation (mm) across the four meteorological seasons, defined as Winter (Dec.–Feb.), Spring (Mar.–May), Summer (Jun.–Aug.), and Autumn (Sep.–Nov.). Among the years examined, 2017 recorded the highest Summer mean rainfall intensity at 0.145 mm/h, accompanied by the highest Summer accumulation of 320.2 mm. By contrast, 2018 was markedly dry, with the lowest Summer intensity (0.069 mm/h) and accumulation (152.4 mm). The highest annual accumulation occurred in 2024 (907.0 mm), while 2018 recorded the lowest (562.6 mm). Overall, rainfall intensity and accumulation vary considerably across seasons and years, with Summer generally contributing substantially to annual rainfall. This seasonal disparity, together with the higher frequency and intensity of convective precipitation during Summer, motivated restricting the training and validation datasets to the summer months.

Table 3.7: Seasonal rainfall intensity and total accumulation across Germany.

Year	Mean Intensity (mm/h)				Total Accumulation (mm)				
	Win.	Spr.	Sum.	Aut.	Win.	Spr.	Sum.	Aut.	Annual
2015	0.063	0.072	0.103	0.089	136.1	159.0	227.4	194.4	716.9
2016	0.074	0.072	0.110	0.065	161.6	159.0	242.9	142.0	705.5
2017	0.043	0.067	0.145	0.097	92.9	147.9	320.2	211.8	772.8
2018	0.079	0.067	0.069	0.042	170.6	147.9	152.4	91.7	562.6
2019	0.082	0.082	0.090	0.094	177.1	181.1	198.7	205.3	762.2
2020	0.092	0.050	0.116	0.066	201.0	110.4	256.1	144.1	711.6
2021	0.079	0.078	0.139	0.055	170.6	172.2	306.9	120.1	769.8
2022	0.077	0.053	0.079	0.090	166.3	117.0	174.4	196.6	654.3
2023	0.069	0.086	0.119	0.109	149.0	189.9	262.8	238.1	839.8
2024	0.108	0.100	0.108	0.097	235.9	220.8	238.5	211.8	907.0

Values represent seasonal averages aggregated from the corresponding yearly radar observations. Abbreviations: Win. = Winter, Spr. = Spring, Sum. = Summer, Aut. = Autumn.

3.3.6 Summary

Several important conclusions derived from the statistical analysis of the radar data influenced the methodological choices made in this study. During most observations, few or no pixels show measurable precipitation (Table 3.5), while strong convective rainfall is rare (Table 3.6). Furthermore, the intensity distribution analysis reveals a strongly right-skewed precipitation distribution with a low median and a small number of extremely high values, indicating occasional intense convective events (Table 3.5 and 3.6). Based on these observations, several preprocessing steps were introduced during data preparation. The radar data were transformed using a logarithmic scale (Section 3.4.3), NaN values were handled (Section 3.4.1), class imbalance was addressed through the sampling strategy (Section 3.5.5), and only months with frequent convective events were selected for dataset construction (Section 3.5).

3.4 Preprocessing

Before model training, the radar and satellite datasets require several preprocessing steps to make them suitable for use in the forecasting model. The raw observations cannot be used directly because the two datasets differ in their spatial representation, statistical distribution, and spatial resolution. The preprocessing procedures are therefore applied to ensure that both datasets have a consistent format for subsequent analysis and model development. The following subsections describe the individual preprocessing steps applied to the radar and satellite data.

3.4.1 Missing Data Handling

The radar and satellite datasets contain two types of missing data, each requiring a different preprocessing strategy. The first concerns missing pixel values within the radar images, while the second concerns missing timestamps between the radar and satellite datasets. Therefore, these two missing-data cases were handled separately before the datasets were used for model training.

In the radar image dataset, about 39–43% of the data contains NaN values, which must be handled carefully (Table 3.4). To address NaN values, files with too many missing values can be removed, or NaN pixels can be replaced with zeros and a mask used to identify them. In this study, the latter approach was adopted. NaN pixels were replaced with zeros and a binary validity mask was created, where 1 indicates a valid pixel and 0 indicates a missing pixel. The mask was used in the loss function so that missing pixels do not contribute to the training loss. Therefore, this approach allows missing observations to be properly handled during model training and evaluation.

The second issue is that the two datasets do not have matching timestamps. Some timestamps are missing due to technical issues during the download. The model requires both data sources to have matching timestamps for metadata creation, analysis, and training. If the input and output timestamps differ, the model could produce incorrect results. To address this issue, only samples with timestamps available in both datasets were retained, ensuring temporal alignment between the two data sources.

3.4.2 Data Normalisation

Scaling the input data is an important preprocessing step for stable neural network training. This is particularly relevant when the data are strongly skewed or contain extreme values. One commonly used approach is Min–Max scaling, which transforms the data into a predefined range, typically $[0, 1]$ [38]. However, when the scaling limits are determined from the observed minimum and maximum values, the method can be sensitive to extreme observations.

Radar precipitation data contain many values close to zero and relatively few high-intensity values. Under Min–Max scaling, intense convective events would occupy the upper end of the scaling range, while most light and moderate precipitation values would be concentrated within a narrow interval. This could reduce the numerical distinction

between the precipitation intensities that occur most frequently. The radar observations used in this study range from 0 to 128 mm/h and are available at five-minute intervals. Therefore, retaining sufficient separation between different precipitation intensities is important for learning both common rainfall patterns and intense convective events.

For this reason, the scaling method was selected separately for each data source. A logarithmic transformation was applied to the radar data to reduce the effect of the wide intensity range while retaining the ordering and relative differences between precipitation values (Section 3.4.3). Z-score normalisation was applied separately to the satellite channels using their global means and standard deviations (Section 3.4.4).

3.4.3 Log Transformation for Radar Data

The radar data were transformed according to a logarithmic formulation designed to map precipitation intensities to a bounded numerical range. The transformation uses a base-10 logarithm together with a noise-floor threshold and a structural offset ϵ to handle zero-valued observations [39]:

$$f(x) = \frac{\log_{10}(\max(x, z) + \epsilon)}{\log_{10}(C_{\max} + \epsilon)} \quad (3.1)$$

Here, z denotes the lower threshold, $\epsilon = 1.0$ is the offset, and $C_{\max} = 128.0$ defines the upper clipping boundary. The inverse transformation is applied during inference to recover precipitation values in their original physical units.

$$f^{-1}(y) = \text{clamp}\left(10^{y \cdot \log_{10}(C_{\max} + \epsilon)} - \epsilon, 0, C_{\max}\right) \quad (3.2)$$

If the reconstructed value is below the noise floor z , it is set to zero to remove any remaining floating-point artifacts.

3.4.4 Z-Score Normalisation for Satellite Data

The satellite has channels CH7 and CH9, and it follows an approximately Gaussian distribution. Z-score normalisation was therefore applied to the satellite data, which shifts each channel to have a mean of zero and a standard deviation of one [40].

$$\hat{x} = \frac{x - \mu}{\sigma + \alpha} \quad (3.3)$$

where μ is the mean of the channel and σ is the standard deviation, and $\alpha = 1 \times 10^{-8}$ is a small regularisation constant that helps prevent division by zero. The channel-specific mean and standard deviation were calculated using the training data and then applied unchanged to the validation and test datasets. This prevents information from the validation and test sets from being used during normalisation.

3.4.5 Reprojection and Resampling of Satellite Imagery

Before the radar and satellite observations could be combined, their different resolutions, image dimensions, and spatial grids had to be matched. Simply resizing the satellite images to the radar dimensions would not be sufficient because pixels at the same image position could represent different geographical locations. Two simple resizing approaches show this problem. Downsampling the higher-resolution data to a lower-resolution grid would combine neighbouring pixels, which can blur the boundaries of convective cells and reduce spatial detail. This can remove small-scale features and change the spatial structure of precipitation cells.

Upsampling the lower-resolution data to a higher-resolution grid also has limitations because it cannot recover details that were not recorded in the original observation [41]. It only estimates new pixel values from the surrounding measurements, increasing the number of pixels without adding new information. As a result, the higher-resolution grid can provide the appearance of additional spatial detail that is not present in the original data. These limitations are important for convective precipitation nowcasting, where the position, boundaries, and shape of precipitation cells provide important information for prediction.

To address this issue, the satellite observations were transformed using their geographical coordinates rather than their original pixel positions [42]. The satellite pixel positions were first converted to geographic coordinates and then transformed into the radar coordinate system. The resulting measurements were mapped onto the fixed radar grid, so that the satellite observations corresponded to the same spatial locations as the radar data. This enabled direct pixel-by-pixel alignment between the two data sources and avoided additional coordinate transformations during later processing.

For each file, the satellite measurements were extracted and interpolated onto the fixed radar grid of 1100×900 cells. Linear interpolation was used, with nearest-neighbour interpolation applied when linear interpolation could not be computed. This produced satellite images with the same dimensions as the radar data and preserved the available spatial information across the entire radar domain. Examples of the regridded satellite data at consecutive time steps are shown in Figure 3.3.

Radar-Satellite (CH7 / CH9) Overlay Comparison

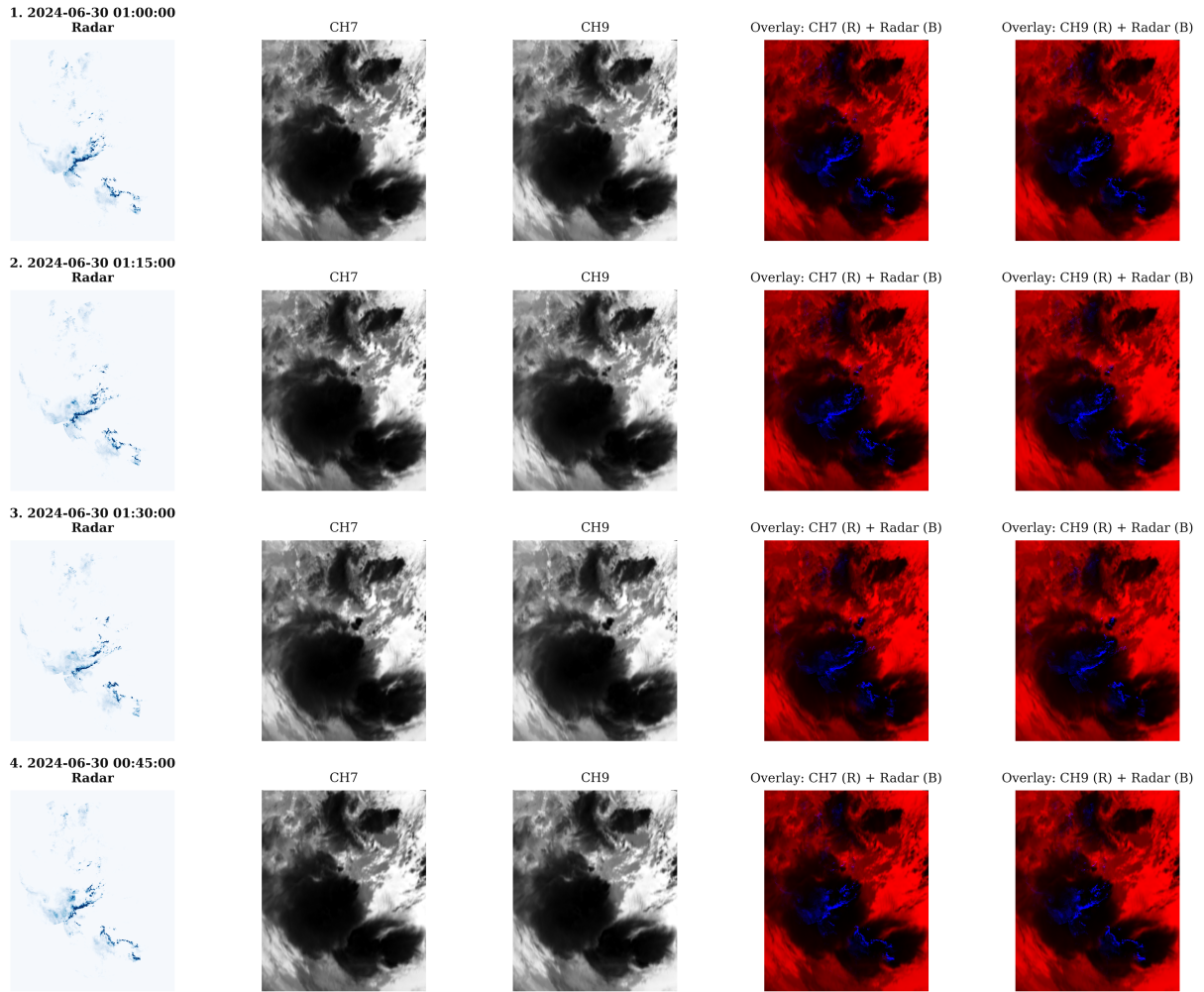


Figure 3.3: Regrided radar and satellite observations over Germany, showing the radar precipitation field and corresponding satellite imagery on a common spatial grid.

The figure confirms that the satellite and radar data are correctly aligned on the same spatial grid. In the overlay images, the satellite observations are represented in ■ red, while the radar precipitation is shown in ■ blue. The ■ black regions correspond to cold cloud structures in the satellite observations where little or no radar precipitation is present. The close spatial correspondence between the ■ precipitation regions and the adjacent ■ cloud structures confirms that the reprojection successfully preserves the spatial relationship between the two data sources.

3.5 Dataset Construction

After preprocessing, the data must be organized into a structure that is suitable for model training and evaluation. This involves selecting representative samples, defining the temporal splits, and preparing the input–output pairs required by the forecasting model. Because the dataset comprises several years of high-resolution radar and satellite obser-

vations, it must also be organized to enable efficient storage and retrieval during training. In addition, the natural imbalance between precipitation and non-precipitation events must be considered to ensure that the model learns meaningful rainfall patterns rather than being dominated by dry conditions. The dataset construction process therefore focuses on creating a well-structured, balanced, and reproducible dataset while preserving the spatial and temporal relationships between the different data sources. The following subsections describe each stage of this process in detail.

3.5.1 Seasonal Filtering

Most convective precipitation occurs during the warmer months of the year. Research shows that deep convective thunderstorms and heavy, short bursts of rain peak between June and September [43]. This is due to strong surface heating, elevated atmospheric moisture, and unstable atmospheric conditions. To make sure the model is trained, validated, and tested using periods when convective events are most common and intense, only data from June, July, August, and September (JJAS) were kept for the training, validation, and test sets. This approach reduces the number of non-convective, stratiform, and dry samples that would otherwise make up most of the dataset in the winter months, which helps the model better learn convective events instead of focusing mainly on the more common dry and non-convective conditions. Using the same seasonal period for all three sets also provides a more consistent basis for model evaluation.

3.5.2 Patch-Based Sampling

Instead of using full reprojected images sized 1100×900 pixels, the model uses a patch-based extraction strategy [44]. Full-resolution images are computationally expensive to process directly and would require substantial GPU memory during training. Patch-based sampling reduces the spatial size of each input while retaining local spatial information relevant to precipitation systems. To provide sufficient spatial coverage and increase the number of training samples, each image is split into fixed-size patches of 256×256 pixels using a sliding window with a stride of 64 pixels, creating an overlap of 192 pixels between adjacent patches.

The 256×256 patch size was chosen over smaller alternatives such as 128×128 for two key reasons. First, convective storm systems usually cover areas larger than 128×128 pixels at $1 \text{ km} \times 1 \text{ km}$ resolution, meaning that a smaller patch would often cut off storm cells at the edges and lose important spatial context. Second, a 256×256 patch provides enough information for the model to capture both the precipitation core, such as cloud-top temperature gradients from the satellite image, and nearby convective precipitation cells from the radar data, which are important for accurate nowcasts. For an image sized 1100×900 pixels, the number of patches per image is computed as:

$$N_{\text{patches}} = \underbrace{\left(\left\lfloor \frac{1100 - 256}{64} \right\rfloor + 1 \right)}_{14} \times \underbrace{\left(\left\lfloor \frac{900 - 256}{64} \right\rfloor + 1 \right)}_{11} = 14 \times 11 = 154 \quad (3.4)$$

3.5.3 Metadata Storage

A structured metadata record is created for each extracted patch and saved in a CSV file. This approach ensures reproducibility, makes stratified sampling possible, and supports post-hoc analysis. Each record includes the observation date and time, the dataset split, and the patch's row and column offset of the top-left corner within the full image, which allows for complete traceability. The record also includes the 99th percentile precipitation value for the patch sequence ($p_{99}(x_{\text{seq}})$), which indicates the intensity of the most extreme rainfall in the patch. This value will be used in later stages for threshold-based sampling and to balance the dataset, which helps address the severe class imbalance (Section 3.5.5). The metadata also includes the timestamps and file paths for the input frames, as well as four future radar composites at 15, 30, 45, and 60-minute intervals as multi-horizon outputs. These details are recorded to support the development of a sequence-based model output.

3.5.4 Dataset Balancing via Metadata-Driven Sampling

Balancing the dataset before training is important for building a reliable machine learning model. If the dataset is imbalanced, the model may focus more on the majority class and pay less attention to the less common classes. In radar data, non-precipitating areas dominate the observations, while precipitation events occur less frequently. Several approaches can be used to balance the data and improve model training. Some common methods include data augmentation [45], generative adversarial networks (GANs) [46], random sampling, and the Synthetic Minority Over-sampling Technique (SMOTE) and its variants [47].

Data Augmentation can create additional data from the existing dataset using techniques such as flipping, colour distortion, and zooming in or out. This can help increase the representation of certain patterns, but it can also change the data distribution and does not fully create the missing information. Generative Adversarial Networks can be used to generate synthetic data for balancing the dataset. GANs create synthetic data using a generator and a discriminator network, but this process is time consuming and requires considerable GPU resources for training. The generated data might also require additional evaluation to ensure that it represents realistic precipitation patterns. In this project, the dataset has over 1 million samples before patching, making the use of GANs for generating additional training data computationally expensive. Class Weighting modifies the loss function to assign higher weights to certain instances rather than simply balancing the classes during training [48]. However, selecting appropriate weights is important for highly skewed radar precipitation data, as excessive weighting of rare high-intensity values may make the model more sensitive to noisy or unrepresentative samples.

Random Oversampling increases the number of minority class instances by randomly duplicating existing samples until the classes are more balanced. However, repeatedly using the same samples can increase the risk of overfitting. Random Undersampling balances the dataset by randomly reducing the majority class, but this may discard useful samples and important precipitation patterns. Synthetic Minority Over-sampling Technique (SMOTE) generates synthetic minority class samples by interpolating between existing samples in feature space. Although this approach can be effective for feature-

based datasets, it is less suitable for the image based radar and satellite data used in this project. Applying SMOTE directly to 256×256 patch based data may not preserve the spatial structures of precipitation fields and could therefore result in unrealistic synthetic samples.

To address this problem, quantile sampling was applied (Section 3.5.5). This method helps balance the training data by including different precipitation conditions. It also provides the model more opportunities to learn from less frequent convective events.

3.5.5 Two-Bucket Quantile Sampling

To ensure that the training data better represent different precipitation conditions, a sampling strategy was applied before model training. A two-bucket quantile sampling approach was used to address the class imbalance. Instead of modifying the data, this approach operates only at the index level using the $p99(x_{\text{seq}})$ field. A quantile threshold τ is calculated from the distribution of $p99(x_{\text{seq}})$ values across all patches in a given split:

$$\tau = Q_{1-r}\left(\{p99(x_i)\}_{i=1}^N\right) \quad (3.5)$$

The parameter r controls the proportion of patches assigned to the upper bucket. Patches are then assigned to one of two buckets:

$$B^+ = \{i : p99(x_i) \geq \tau \text{ and } p99(x_i) > 0\} \quad (3.6)$$

$$B^- = \{i : p99(x_i) < \tau \text{ or } p99(x_i) = 0\} \quad (3.7)$$

Completely dry patches are always placed in B^- , even if the threshold drops to zero. Each patch is kept or discarded based on its bucket, where $u_i \sim \mathcal{U}(0, 1)$:

$$\text{keep}(i) = \begin{cases} u_i < p & \text{if } i \in B^+ \\ u_i < 1 - p & \text{if } i \in B^- \end{cases} \quad (3.8)$$

The parameter p determines the probability of retaining patches from the upper bucket, while patches in the lower bucket are retained with probability $1 - p$. A fixed random seed is used for each split to ensure reproducibility. For this study, the quantile threshold was set to $r = 0.3829$ and the retention probability to $p = 0.6171$, corresponding directly to the empirical proportion of training patches containing measurable precipitation ($p99(x_{\text{seq}}) > 0$). As $p = 1 - r$, this provides approximately 1:1 balance between the top and rest buckets. Both parameters can be adjusted by the user through command-line arguments to explore different sampling configurations.

This approach has a few practical benefits. The radar and satellite data are not modified or deleted, as sampling is applied only at the metadata level. This means that excluded patches remain available if a different sampling configuration is needed later. No synthetic data are generated. The pre sampled CSV file is directly used by the data loader without additional resampling. Sampling is applied to the training and validation sets, while the test set is retained for final evaluation.

3.6 Input and Output Sequence Definition

The input and output sequences are defined according to the sampling characteristics of the available observations to provide a consistent structure for model training and evaluation. The selected intervals determine how recent observations are combined and how future precipitation is represented in the forecasting task. The input and output sequences are paired to form individual training examples, allowing the deep-learning model to learn the relationship between past observations and their corresponding future precipitation fields.

3.6.1 Input Sequence

The model takes in a short sequence of recent observations as its input. At each reference time t , the input sequence includes $N_{in} = 4$ consecutive frames, each representing a 5-minute precipitation observation period, together covering the most recent 20 minutes of input precipitation information:

$$\mathcal{X}_t = \{x_t^{(1)}, x_t^{(2)}, x_t^{(3)}, x_t^{(4)}\}, \quad (3.9)$$

where $x_t^{(i)}$ denotes the i -th frame in the input sequence, arranged in chronological order, with each frame corresponding to a distinct 5-minute observation period. Each input frame has two satellite channels (CH7 and CH9) and one radar precipitation field. This gives each frame three feature channels across the 256×256 spatial grid.

The 20-minute input window is suitable for short-range convective nowcasting, where a compact but informative history can capture the recent movement of precipitation systems while also keeping the model input compact and computationally tractable [8].

3.6.2 Output Sequence

The model is designed to predict the radar precipitation field at four different future times. These prediction times are spaced 15 minutes apart, with predictions made up to 60 minutes ahead:

$$\mathcal{Y}_t = \{y_t^{(1)}, y_t^{(2)}, y_t^{(3)}, y_t^{(4)}\} \quad (3.10)$$

where $y_t^{(i)}$ denotes the radar precipitation field at the i -th forecast lead time. The four prediction targets correspond to lead times of 15, 30, 45, and 60 minutes relative to the reference time t .

The four-step multi-output setup allows the model to predict precipitation at multiple future horizons rather than only a single future state. A maximum lead time of 60 minutes was chosen as it is a common forecast horizon used in precipitation nowcasting [49]. Predictions were made at 15-minute intervals to evaluate how forecast accuracy changes across the prediction horizon. The full input and output sequence configuration used in this study is shown in Table 3.8.

Table 3.8: Input and output sequence configuration.

Property	Input	Output	Value
Frames	$N_{in} = 4$	$N_{out} = 4$	4
Frame length	5 min	—	—
Lead times	—	15, 30, 45, 60 min	—
Window	20 min	60 min	—
Channels	Radar + CH7 + CH9	Radar only	3 / 1
Spatial size	256 × 256 pixels		

Four input frames, each covering a 5-minute observation period, together represent a 20-minute observation sequence. The output sequence consists of four predictions at 15-minute intervals, up to a 60-minute forecast horizon.

3.6.3 Dataset Partitioning

The dataset was divided into three non-overlapping subsets: training, validation, and test. The training set was used to optimize model parameters, the validation set to monitor generalisation and select the best-performing configuration, and the test set was reserved for final evaluation. Since the meteorological observations are continuous and ordered in time, temporal splitting was adopted [50]. This approach preserves the temporal order and prevents future observations from entering the training process, thereby reducing the risk of data leakage and overly optimistic evaluation. The dataset was therefore partitioned into distinct time periods for training, validation, and testing, as shown in Tables 3.9 and 3.10.

Table 3.9: Temporal dataset split – training and validation sets.

Split	Period	Duration	Before	After
Training	2015–2022	8 years	27,197,532	12,851,153
Validation	2023	1 year	3,377,596	1,586,763

The sample counts correspond to the JJAS (June–September) period used for dataset construction, at the patch level (256×256), consistent with the patch-based model input. The “Before” column shows the total number of valid patch samples generated before sampling, while the “After” column shows the number of samples retained after the two-bucket quantile sampling procedure (Section 3.5.5).

Table 3.10: Temporal dataset split – test set.

Split	Period	Duration	Samples
Test	2024	1 year	32,021

The 2024 test set was used for the final performance evaluation and was not subjected to the two-bucket quantile sampling procedure (Section 3.5.5). The reported count represents the complete set of valid JJAS timepoints. Unlike the training and validation sets, the test count is reported at the full-domain level for each timepoint, consistent with the full-domain sliding-window inference procedure used for evaluation, rather than at the patch level.

The models were trained on fixed-size 256×256 patches. Since a trained model expects this fixed input size, it cannot directly accept a full-size image as input. To address this, a sliding-window approach was used. The full image was divided into overlapping 256×256 patches, each processed independently by the model, and the resulting predictions were recombined using cosine-weighted blending to obtain a single full-domain output. This allowed the evaluation to be performed over the complete spatial domain, reflecting realistic deployment conditions rather than isolated patches.

3.7 Convective Event Definition

Before constructing the training dataset, a clear and objective definition of a convective event is required. Convective precipitation is typically associated with localized, short-lived, and high-intensity rainfall structures. Radar reflectivity thresholds are commonly used for convective rainfall, with values above 40 dBZ generally considered indicative of convective activity [51]. However, the RADOLAN dataset used in this study provides precipitation rates rather than radar reflectivity [36]. Therefore, reflectivity-based criteria could not be applied directly, and convective events were identified using a two-level storm detection method based on precipitation rates. The corresponding detection criteria are described below.

3.7.1 Identification of Convective Events

An operational threshold and a higher extreme threshold were used to identify low-level convective cells and the most intense convective cores, respectively. These thresholds were set to 15 mm/h and 40 mm/h, based on the Deutscher Wetterdienst's (DWD) [52] official 1-hour Starkregen warning criteria for markante Wetterwarnung and extreme Unwetterwarnung, respectively.

First, the input rainfall data are cleaned by removing invalid values and retaining only valid pixels. The operational threshold is then applied to identify pixels with rainfall at or above the threshold. A morphological closing operation with a structuring element size of 4 pixels is applied to the resulting mask. This size was chosen empirically, as smaller sizes may not fill internal gaps and can cause convective regions to break into smaller fragments, while larger sizes can over-smooth the data and merge separate convective cells. Connected component labelling is then applied, so each connected region is treated as a candidate convective event.

The extreme threshold is then applied within the operational convective event regions to identify pixels with extreme rainfall intensities. The resulting extreme mask is processed in the same way using morphological closing and small object removal. The complete detection procedure was then applied to the model predictions to annotate the corresponding regions on the predicted precipitation fields.

Chapter 4

Methodology and Implementation

Convective precipitation nowcasting involves several stages, ranging from data preparation to model evaluation. This chapter explains how each stage of the proposed framework was implemented. Figure 4.1 summarizes the overall workflow for the multimodal configuration. Radar and satellite observations are first collected and prepared before being merged into a single multimodal representation using the selected fusion method (Section 4.2.1). The fused data are subsequently used as input to the five deep learning models (Section 4.2). Each model produces radar precipitation forecasts for four prediction horizons.

Model training is performed using the hybrid loss function (Section 4.3.1), which is designed to improve the forecasting of precipitation fields across different prediction horizons. The predicted radar fields are then assessed using the selected evaluation metrics (Section 4.4 and its subsections). Together, these metrics quantify forecast accuracy as well as the structural quality of the predicted images. The chapter concludes with a description of the experimental settings, including the optimization procedure, training configuration, and computational hardware used for all experiments.

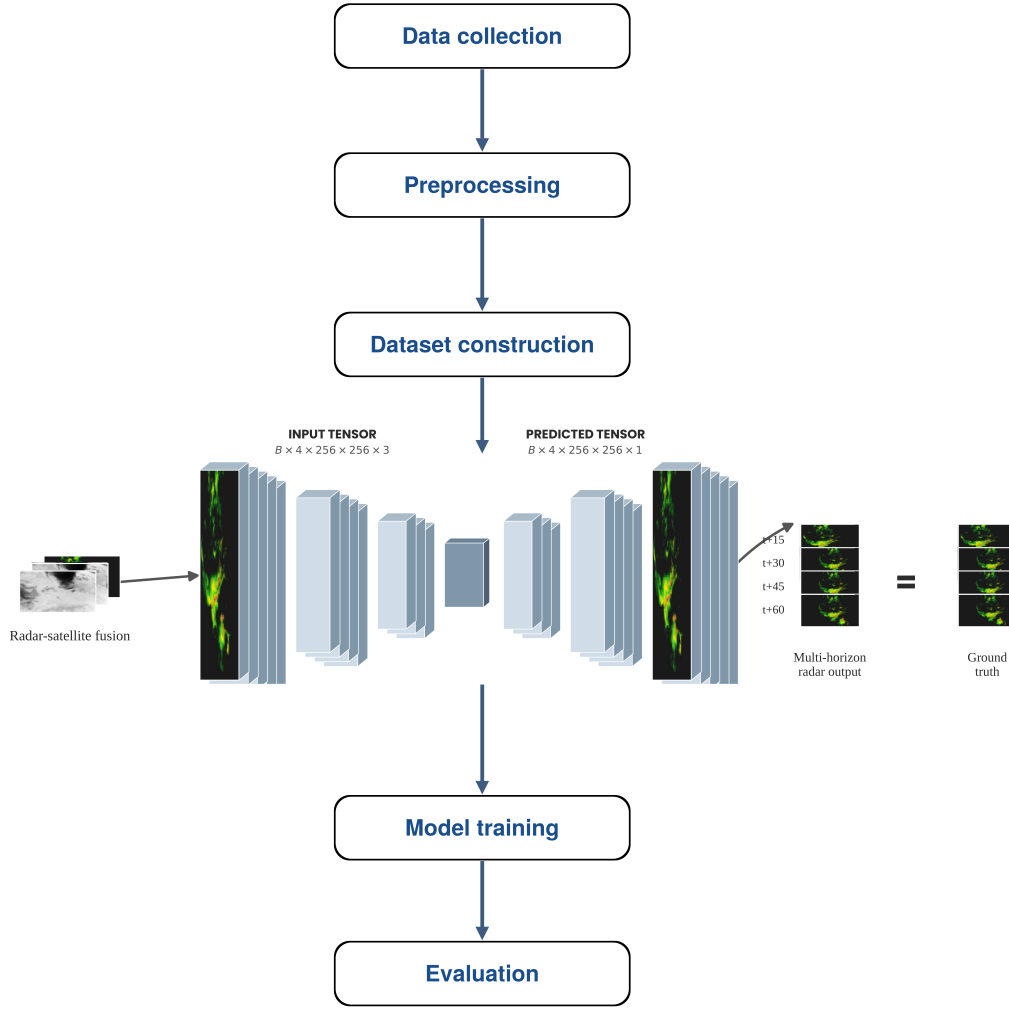


Figure 4.1: Overview of the proposed convective precipitation nowcasting methodology, showing the complete processing pipeline and the flow of data through the different stages. The multimodal configuration combines radar and satellite observations, while the radar-only configuration follows the same pipeline with the satellite input removed.

4.1 Sequence-to-Sequence Forecasting Problem

Convective precipitation nowcasting is formulated as a sequence-to-sequence prediction problem [53]. The objective is to learn a mapping from a sequence of recent multimodal observations to a sequence of future radar precipitation fields over the forecasting horizon. This formulation is well suited to convective precipitation because storm systems evolve continuously in both space and time. Rather than predicting a single future state, the model learns the temporal evolution of precipitation across multiple forecast lead times. Therefore, the forecasting model is defined as

$$f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} \quad (4.1)$$

where X denotes the input sequence of multimodal observations, $\mathcal{Y}$ represents the corresponding sequence of future radar precipitation fields, and f_θ is the forecasting model parameterized by θ . Radar precipitation is selected as the prediction target because it provides observations of surface precipitation and serves as the operational reference for precipitation nowcasting [54].

The forecasting problem is addressed as a supervised learning task as well, where each input sequence is paired with its corresponding future radar sequence. During training, the model parameters are optimized by minimizing the difference between the predicted and observed radar precipitation fields. The resulting model is expected to produce forecasts that accurately represent the underlying precipitation patterns and their variations throughout the entire prediction horizon.

4.2 Model Architecture

This study trains and compares several deep learning models for predicting convective precipitation. Each model is built from the ground up, using the same input and output setup described in (Section 3.6). The following subsections first describe how the radar and satellite inputs are fused, and then present each of the individual architectures.

4.2.1 Input Fusion Strategy

Radar and satellite data acquired from different sources, so these two sources need to be combined before being used in the model. The most common and straightforward way to do this is channel concatenation [30]. In this method, radar and satellite channels are stacked along the channel dimension to form a single input tensor. This allows the model to access all three channels simultaneously and learn to use them together. At each timestep t , the radar field and the two satellite channels are combined along the channel axis:

$$\underbrace{(256 \times 256 \times 1)}_{\text{Radar}} + \underbrace{(256 \times 256 \times 2)}_{\text{CH7, CH9}} \xrightarrow{\text{concatenate}} \underbrace{(256 \times 256 \times 3)}_{\text{Input frame}}$$

The full input sequence is created by stacking four consecutive frames:

$$\underbrace{4 \times (256 \times 256 \times 3)}_{\text{4 timesteps}} \xrightarrow{\text{stack in time}} \underbrace{4 \times 256 \times 256 \times 3}_{\text{Model input}}$$

This method is easy to implement and does not add extra parameters to the model. For this reason, simple channel concatenation was used for all five models.

4.2.2 ConvLSTM

ConvLSTM [16] combines the convolution operation with the LSTM structure. The model receives a sequence of precipitation observations and processes the frames one time step at a time. At each step, convolution is used to extract spatial features, while the hidden and cell states carry information from the previous time steps. The input, hidden state, and cell state are updated through the LSTM gates, allowing the model to decide which information should be retained or updated. The updated hidden state is then used when processing the next frame. For precipitation nowcasting, this allows the model to use the recent movement and development of precipitation patterns when predicting future frames.

4.2.3 SmaAt-UNet

SmaAt-UNet [5] follows an encoder-decoder structure similar to U-Net. The input precipitation fields first pass through the encoder, where their spatial dimensions are reduced while higher-level features are extracted. The decoder then gradually increases the spatial resolution to reconstruct the predicted precipitation field. Information from the encoder is passed to the decoder through skip connections, helping to retain details that may be lost during downsampling. SmaAt-UNet uses depthwise separable convolutions to reduce the number of parameters and attention gates in the skip connections to give more weight to important regions. This is useful for the present task because convective cells can occupy only a small part of the radar field and their spatial details are important for accurate prediction.

4.2.4 Earthformer

Earthformer [6] uses a Transformer-based structure to process spatiotemporal data. The input sequence is divided into smaller 3D cuboids that contain information from both the spatial dimensions and the time dimension. The model applies self-attention to these cuboids so that information from different locations and time steps can be related to each other. The resulting features are passed through the Transformer layers to build a representation of the spatiotemporal patterns in the input sequence. The model then uses this representation to predict the future frames. For precipitation nowcasting, this approach is useful because the development of a convective cell at one location can be related to changes occurring in other locations and at earlier time steps.

4.2.5 SimVP

SimVP [55] uses a convolutional encoder and decoder to predict future frames from a sequence of observed frames. The encoder first converts the input frames into feature representations, which contain information about the observed precipitation patterns. These features are then processed by the prediction component to learn the changes between the observed and future states. Finally, the decoder converts the learned features back into the spatial form of the predicted precipitation fields. SimVP does not use

recurrent connections or attention mechanisms, so the model provides a simpler way of learning the relationship between past and future precipitation fields. This makes it useful in this study as a convolution-based model for comparison with the recurrent and Transformer-based architectures.

4.2.6 VPTR

VPTR [56] combines a convolutional encoder with a Transformer decoder. The input sequence is first passed through the convolutional encoder, which extracts spatial features from the radar and satellite observations. These features are then provided to the Transformer decoder, where the attention mechanism processes relationships within the encoded sequence. The decoder uses this information to generate the future frames. In this way, the convolutional encoder handles the extraction of local spatial features, while the Transformer is used to model relationships across the sequence. This combination is suitable for convective precipitation nowcasting because the prediction requires both the detailed spatial structure of convective cells and information about how these structures change over time.

4.2.7 Ensemble Model

In addition to the five models, an ensemble modeling technique has been incorporated by considering predictions of all five models [57]. The ensemble prediction for each forecast horizon is the arithmetic mean of the five model predictions:

$$\hat{y}_{\text{ensemble}} = \frac{1}{5} \sum_{i=1}^5 \hat{y}_i \quad (4.2)$$

where $\hat{y}_i$ is the prediction of model i . Taking the mean of all five models may reduce the influence of errors associated with individual architectures, depending on the degree of diversity between their predictions.

4.2.8 Summary

The architectures evaluated in this study include recurrent, convolutional, and Transformer-based models. In the multimodal configuration, each architecture receives an input of shape (4, 3, 256, 256), corresponding to four consecutive frames with radar, CH7, and CH9 channels, and produces an output of shape (4, 256, 256) covering the four forecast horizons. The radar-only configuration uses the same temporal sequence and spatial dimensions, with only the radar channel provided as input. The details of this configuration and its comparison with the multimodal setup are described in the Radar-Only Input section (Section 5.2).

4.3 Loss Function

A loss function measures the difference between the model's predictions and the target values. During training, it provides feedback on the prediction error and guides the update of the model parameters. For image-to-image regression, the prediction and target are spatial fields, so the error is calculated between corresponding pixel locations and then combined into an overall loss value. A lower loss indicates better agreement between the predicted and observed values. The choice of loss function is important because different formulations give different importance to prediction errors and therefore affect the training of the model.

Several loss functions are commonly used for image-to-image regression, including Mean Squared Error (MSE) [58], Mean Absolute Error (MAE) [59], and Root Mean Squared Error (RMSE) [60]. However, these standard losses do not explicitly account for the uneven distribution of precipitation intensities in radar observations. When the loss is calculated uniformly across pixels, errors associated with the less frequent high-intensity precipitation values may have limited influence on the overall training objective. This is particularly important for precipitation nowcasting, where accurately representing intense rainfall events is an important part of the forecasting task.

To address the limitations of standard loss functions for precipitation forecasting, we use a hybrid loss function based on weighted Mean Absolute Error (MAE). Pixel-wise errors are calculated and averaged over the valid regions. A user-defined multilevel weighting scheme assigns greater weights to larger precipitation values, giving greater emphasis to heavy rainfall events. The details of the weighting scheme and the proposed loss formulation are presented in (Section 4.3.1).

4.3.1 Hybrid Loss Function

To address the limitations described above in (Section 4.3), this project used a five-tier intensity-weighted approach. The idea was to give more importance to pixels with higher precipitation intensities so that the model could focus more strongly on intense convective events during training [48]. The pixel weights are determined from the target precipitation intensity (mm/h) after applying the inverse log transformation. A base weight of 1.0 is assigned to dry and light-rain pixels, while higher weights are assigned as the precipitation intensity reaches the defined thresholds.

Weighted Mean Absolute Error (MAE) was selected over Mean Squared Error (MSE) because MAE is less sensitive to large individual errors. This is useful for the precipitation data used in this study, where high-intensity rainfall values are relatively rare. The weighted MAE is given by:

$$\mathcal{L}_{\text{MAE}} = \frac{1}{|\mathcal{M}|} \sum_{(x,y) \in \mathcal{M}} w(x,y) |\hat{y}_{x,y} - y_{x,y}| \quad (4.3)$$

Here, $\hat{y}_{x,y}$ denotes the predicted value and $y_{x,y}$ denotes the target value at pixel (x,y) . The term $w(x,y)$ represents the corresponding precipitation-based pixel weight, while $\mathcal{M}$

contains the valid pixels identified by the NaN mask described in (Section 3.4). Pixels marked as NaN are excluded from the calculation so that invalid radar values do not contribute to the training loss.

The precipitation thresholds and weights follow a weighted-loss formulation [1], assigning higher weights to more intense rainfall. A five-tier scheme uses thresholds of 3, 7, 15, 25, and 40 mm/h, with weights equal to the thresholds. The 15, 25, and 40 mm/h thresholds correspond to the DWD [52] Starkregen warning criteria, while 3 and 7 mm/h represent lower intensities. The scheme is shown in Table 4.1.

Table 4.1: Intensity-weighted threshold scheme for spatial loss calculation.

Tier	Threshold (mm/h)	Weight
1	≥ 3.0	3.0
2	≥ 7.0	7.0
3	≥ 15.0	15.0
4	≥ 25.0	25.0
5	≥ 40.0	40.0

The thresholds and weights are defined in mm/h after the inverse log transformation, with weights assigned according to precipitation intensity.

In addition to the intensity weighting, a gradient-sharpness loss was included to preserve the spatial characteristics of the predicted precipitation fields. It compares the spatial gradients of the predicted and target fields, encouraging sharper precipitation structures and reducing excessive smoothing. The contribution of this component is controlled by the parameter λ . In this study, λ was set to 0.2 and kept fixed for all models.

The model produces four future radar frames, with the loss calculated for each horizon and combined using different temporal weights. Later horizons receive larger weights to place greater emphasis on the longer-term forecast performance. The selected temporal weights are given in Table 4.2.

Table 4.2: Temporal horizon weights for multi-step forecast optimization.

Horizon	Lead Time (min)	Weight w_h
1	15	1.0
2	30	1.5
3	45	2.0
4	60	2.5

The weights are assigned to the individual forecast horizons, with larger values given to predictions at longer lead times.

For each forecast horizon h , the intensity-based pixel weighting and the gradient-sharpness term are incorporated into the loss:

$$\mathcal{L}_h = \frac{1}{|\mathcal{M}|} \sum_{(x,y) \in \mathcal{M}} w(x,y) |\hat{y}_{x,y,h} - y_{x,y,h}| + \lambda \mathcal{L}_{\text{grad},h} \quad (4.4)$$

where $\mathcal{L}_{\text{grad},h}$ denotes the gradient-sharpness loss for horizon h . The parameter $\lambda = 0.2$ controls the contribution of the gradient term relative to the intensity-weighted MAE and was kept fixed throughout the experiments. The four horizon losses are combined using their respective temporal weights:

$$\mathcal{L}_{\text{total}} = \sum_{h=1}^4 \bar{w}_h \mathcal{L}_h \quad (4.5)$$

where $\mathcal{L}_h$ denotes the weighted MAE with the gradient-sharpness term. The normalised horizon weight $\bar{w}_h$ is obtained by dividing the raw weight for horizon h by the sum of the four horizon weights.

4.4 Model Evaluation Metrics

To evaluate the performance of the proposed models, five evaluation metrics are employed. The evaluation is conducted on the test dataset (Section 3.6.3), which contains unseen samples and therefore provides an independent assessment of the models' generalization capability. The selected metrics are Mean Absolute Error (MAE), Mean Squared Error (MSE), Critical Success Index (CSI), Equitable Threat Score (ETS), and Peak Signal-to-Noise Ratio (PSNR). Lower values indicate better performance for MAE and MSE, whereas higher values indicate better performance for CSI, ETS, and PSNR. For consistency, the same set of metrics is also applied to the non-deep-learning baseline, allowing its results to be compared with those of the deep-learning models under the same evaluation criteria.

4.4.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) is a regression metric that measures the average magnitude of the errors between predicted and observed values [59]. For precipitation nowcasting, it quantifies the average difference between predicted and observed precipitation intensities across the radar field, treating overestimation and underestimation equally.

$$\text{MAE} = \frac{\sum_{i=1}^N |y_i - \hat{y}_i|}{N} \quad (4.6)$$

where y_i and $\hat{y}_i$ denote the observed and predicted precipitation intensities for the i -th pixel, respectively, and N is the total number of pixels.

4.4.2 Mean Squared Error (MSE)

Mean Squared Error (MSE) is a regression metric that measures the average of the squared differences between predicted and observed precipitation intensities [61]. Unlike MAE, MSE gives greater weight to larger prediction errors, making it more sensitive to substantial discrepancies in precipitation intensity and therefore useful for identifying cases where large prediction errors occur.

$$\text{MSE} = \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{N} \quad (4.7)$$

The terms y_i , $\hat{y}_i$, and N have the same meanings as in (Section 4.4.1).

4.4.3 Critical Success Index (CSI)

The Critical Success Index is widely used to evaluate precipitation nowcasting because it measures the ability of a model to correctly identify precipitation events above a specified threshold while ignoring correct negatives [62]. This makes the metric particularly suitable for datasets in which non-precipitating pixels greatly outnumber precipitating pixels. The CSI is computed as

$$\text{CSI} = \frac{H}{H + M + FA} \quad (4.8)$$

where H , M , and FA denote the numbers of hits, misses, and false alarms, respectively. A hit occurs when precipitation is present in both the prediction and observation, a miss occurs when precipitation is observed but not predicted, and a false alarm occurs when precipitation is predicted but not observed. These quantities are determined by comparing the predicted and observed precipitation fields at the selected precipitation threshold.

4.4.4 Equitable Threat Score (ETS)

The Equitable Threat Score extends the CSI by correcting for hits that occur by chance, making it a more reliable metric when precipitation events are infrequent and non-precipitating pixels dominate the dataset [63]. The ETS is computed as

$$\text{ETS} = \frac{H - H_c}{H + M + FA - H_c} \quad (4.9)$$

where H , M , and FA have the same definitions as in the CSI formulation (Section 4.4.3), and H_c denotes the number of hits expected by chance. It is computed as

$$H_c = \frac{(H + M)(H + FA)}{N} \quad (4.10)$$

where $N = H + M + FA + CN$ is the total number of samples, and CN denotes the number of correctly predicted non-precipitation events.

4.4.5 Peak Signal-to-Noise Ratio (PSNR)

Peak Signal-to-Noise Ratio is an index showing how accurately the predicted precipitation pattern fits the actual observation [64].

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \quad (4.11)$$

Here, MAX is the maximum precipitation amount in the data and MSE is the mean squared error between the true precipitation fields and the predicted fields (Section 4.4.2). The base-10 logarithm is used because the error ratio varies widely with precipitation intensity, and a logarithmic scale makes the value easier to interpret and compare across models and datasets.

4.4.6 Summary

Considering multiple evaluation metrics provides a more comprehensive assessment of model performance in precipitation nowcasting. A model may achieve small pixel-wise errors while still failing to accurately capture the location or intensity of precipitation events. The selected metrics are complementary, capturing numerical error, event detection, and overall image quality. Together, these measures provide a broader assessment of the predicted precipitation fields across different forecast lead times.

4.5 Experimental Setup

A consistent experimental setup was used for all five architectures, with the same data splits and hardware and only minor architecture-specific modifications where required. Hyperparameters were refined through multiple experiments using validation performance as the selection criterion. Early stopping, mixed-precision training, and gradient clipping were used to improve training stability and reduce overfitting. The main training and optimisation settings are summarised in Table 4.3.

Table 4.3: Training configuration.

Parameter	Value
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$)
Learning rate	1×10^{-4}
Weight decay	1×10^{-3}
Gradient clipping	0.5
LR schedule	Cosine annealing ($\eta_{min} = 5 \times 10^{-6}, T_{max} = 50$)
Batch size	16
Accumulation steps	2
Max epochs	50
Early stopping patience	30 epochs
Mixed precision	Yes
Data workers	24 (prefetch factor 4)
Hardware	4 × NVIDIA Tesla V100-PCIE (32 GB)
Framework	PyTorch
Containerisation	Docker
Hyperparameter search	Grid Search

The table defines the default training and optimization settings used throughout the study.

For each architecture, the model with the highest ETS on the validation set was selected. ETS was chosen because it accounts for hits expected by chance and is particularly suitable for the imbalanced precipitation data described in (Section 3.5.4). This provides a better and more reliable basis for selecting models for precipitation detection.

Chapter 5

Results

This chapter presents the evaluation results of the five deep learning architectures, the ensemble model, and the non-deep-learning baseline on the held-out test set. The models are compared across all four forecast horizons using the evaluation metrics presented in (Tables 5.1 and 5.2). The results are organized according to the two input configurations considered in this study. The first section evaluates the multimodal models using radar and satellite observations as input, while the second examines the radar-only models. Each section presents a visual comparison of the predicted precipitation fields against the corresponding ground truth, followed by a quantitative evaluation based on the selected performance metrics. Comparing the two input configurations under the same forecasting setup enables assessment of the impact of incorporating satellite observations. In addition, it provides insight into how different model architectures behave as forecast lead times increase and precipitation intensities vary. Alongside these comparisons, a separate baseline evaluation provides a non-deep-learning reference for assessing the deep learning models. Together, these results provide the basis for a detailed interpretation of the observed trends and the limitations of the proposed approach.

5.1 Multimodal Input: Radar and Satellite

Figure 5.1 shows the precipitation forecasts generated from radar and satellite observations for a representative convective event on 30 June 2024. The event features a convective system with precipitation spread across a large part of the domain, including several active cells with high rainfall intensities visible in the ground truth. Over the forecast window, the system gradually evolves, with the most intense cells weakening and parts of the precipitation area shifting position between $t+15$ and $t+60$. The rows represent the individual models and the ensemble, while the columns show the four forecast lead times of 15, 30, 45, and 60 minutes. The aggregated quantitative results across the entire test set and all forecast lead times are presented in Table 5.1.

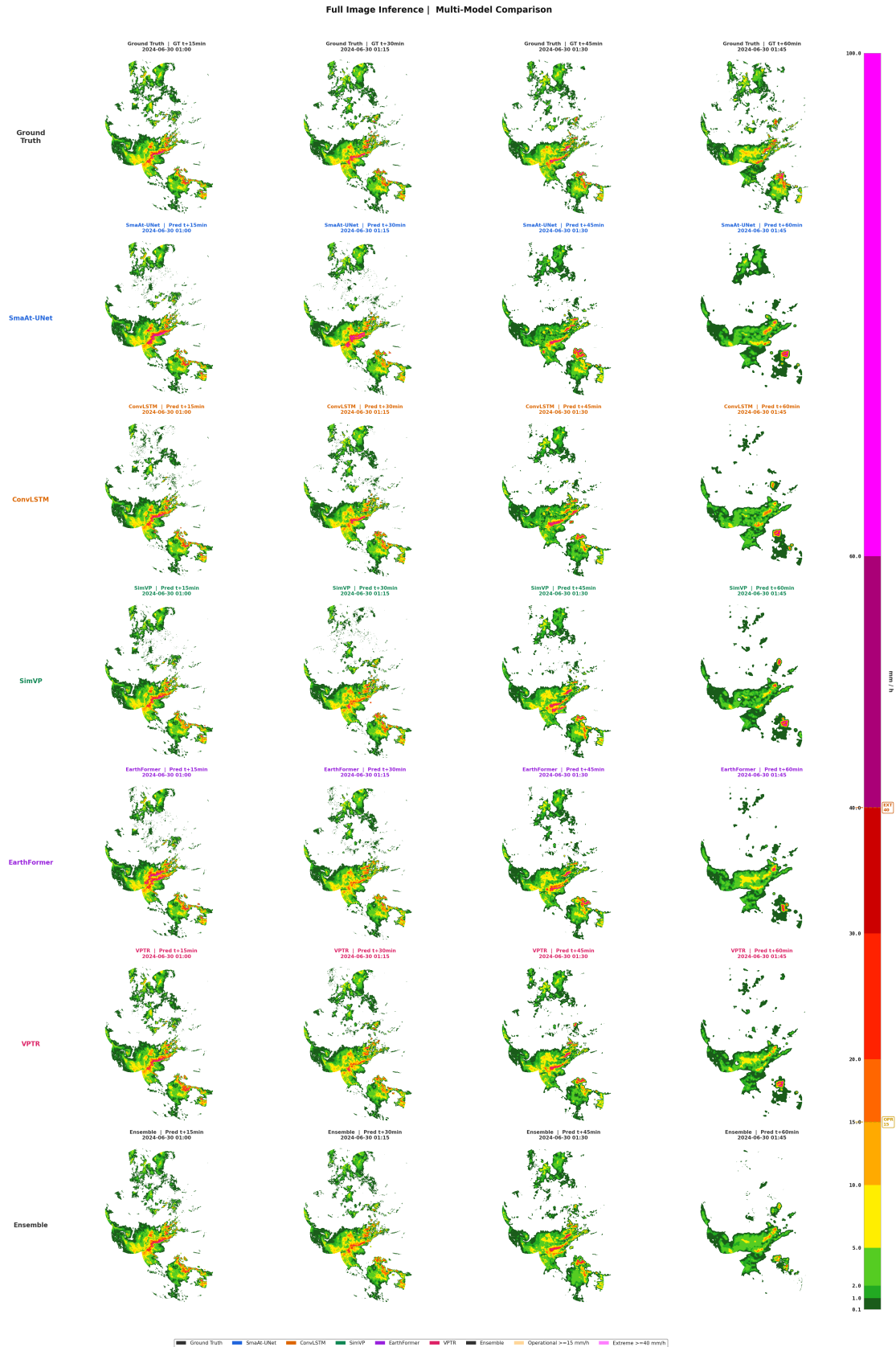


Figure 5.1: Multi-model precipitation forecasts from fused radar and satellite observations for a convective event on 30 June 2024. The event was selected as a representative case for qualitative illustration rather than a systematic evaluation of model performance. The first row shows the radar ground truth at each forecast horizon. Rows correspond to individual models and the ensemble, and columns correspond to forecast horizons of 15, 30, 45, and 60 minutes ahead. Colour indicates precipitation intensity from light rainfall (■) to extreme convective precipitation (■), with the operational threshold marked on the colour bar.

Table 5.1: Performance comparison across multimodal models and lead times.

Lead Time	Model	ETS $\uparrow$	CSI $\uparrow$	MAE $\downarrow$	MSE $\downarrow$	PSNR $\uparrow$
$t + 15$ min	ConvLSTM	0.24	0.28	0.21	2.20	38.7
	EarthFormer	0.28	0.29	0.18	2.00	39.3
	SimVP	0.29	<u>0.32</u>	0.18	1.90	39.5
	SmaAt-UNet	<u>0.27</u>	<u>0.28</u>	0.16	<u>1.80</u>	39.6
	VPTR	0.31	0.33	<u>0.16</u>	1.70	39.8
	Ensemble	0.28	0.30	0.15	1.90	39.4
$t + 30$ min	ConvLSTM	0.21	0.25	0.22	2.23	38.6
	EarthFormer	0.22	0.26	0.22	2.30	38.5
	SimVP	0.25	0.28	0.19	2.00	39.1
	SmaAt-UNet	0.19	0.22	<u>0.24</u>	<u>2.60</u>	<u>38.0</u>
	VPTR	<u>0.24</u>	<u>0.27</u>	0.20	1.92	39.3
	Ensemble	0.22	<u>0.26</u>	0.18	2.21	38.7
$t + 45$ min	ConvLSTM	0.15	0.19	0.27	3.00	37.4
	EarthFormer	0.17	0.20	0.25	2.70	37.8
	SimVP	0.19	<u>0.21</u>	0.24	2.50	38.2
	SmaAt-UNet	<u>0.16</u>	<u>0.18</u>	0.29	<u>2.65</u>	<u>37.9</u>
	VPTR	0.20	0.22	0.22	2.40	38.3
	Ensemble	0.18	0.21	<u>0.23</u>	2.65	37.8
$t + 60$ min	ConvLSTM	0.12	0.13	0.28	3.12	37.2
	EarthFormer	0.13	0.15	0.29	3.10	37.4
	SimVP	0.14	0.16	0.26	2.80	37.7
	SmaAt-UNet	<u>0.11</u>	<u>0.12</u>	<u>0.30</u>	<u>2.81</u>	<u>37.6</u>
	VPTR	0.15	0.17	0.25	2.70	37.8
	Ensemble	0.13	0.15	0.28	2.93	37.5

Bold indicates the best value for each metric at each lead time, while underlining indicates the second-best value. $\uparrow$ indicates that higher values are better, and $\downarrow$ indicates that lower values are better. Tied values receive the same formatting. All CSI and ETS values are averaged across the 5 and 15mm/h thresholds; the subscript is omitted from the column header due to space constraints.

To better understand the quantitative results, Figures 5.2 and 5.3 provide a visual comparison of the multimodal model performance across the four forecast lead times, complementing the numerical results reported in Table 5.1. Together, these figures allow a clearer comparison of the relative performance of the evaluated architectures.

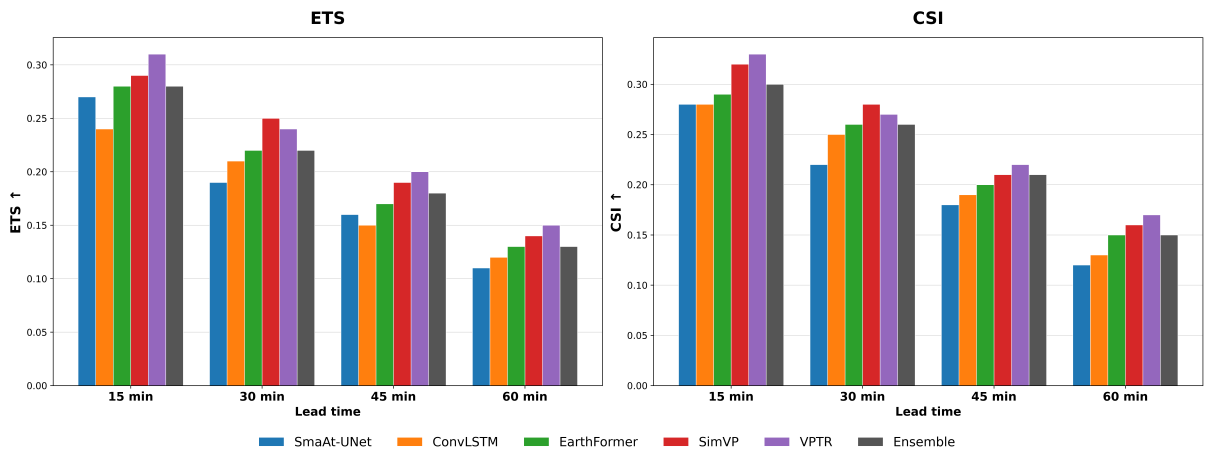


Figure 5.2: Multimodal categorical performance across forecast lead times.

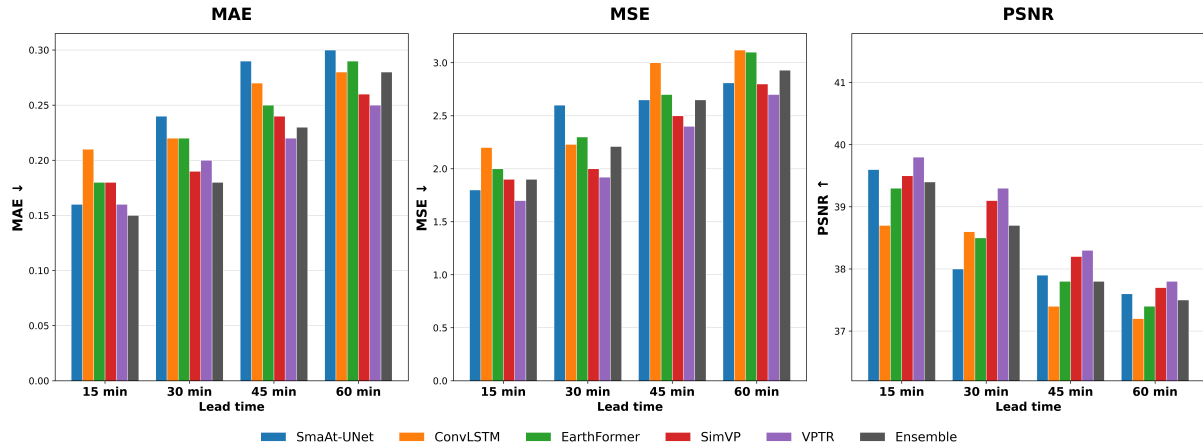


Figure 5.3: Multimodal continuous and image-based performance across lead times.

All models successfully capture the main precipitation region at $t + 15$, showing that the forecasts represent the observed spatial pattern well at the shortest lead time, although some differences in intensity and spatial extent are visible in Figure 5.1. As the forecast lead time increases, the predicted precipitation patterns gradually diverge from the ground truth in both spatial distribution and intensity. Nevertheless, the main precipitation structure remains recognisable in several predictions even at longer lead times. The differences are most noticeable at $t + 60$, when some models fail to reproduce smaller observed precipitation regions. This visual change in the predicted fields corresponds to the variation across lead times shown in Figures 5.2 and 5.3. Overall, the models produce meaningful forecasts throughout the evaluated 60-minute period, while the general decrease in accuracy with increasing lead time is also reflected in the test-set results reported in Table 5.1.

5.2 Radar-Only Input

An additional set of experiments was conducted using only radar observations as model input while preserving the same experimental setup as the multimodal configuration. The two satellite channels, CH7 and CH9, were removed at the input level, reducing each frame from three channels to a single radar precipitation channel while preserving the same spatial grid and temporal sequence. The two input configurations can be expressed as

$$\mathbf{X}_{\text{multi}} = [\mathbf{X}_{\text{radar}}, \mathbf{X}_{\text{CH7}}, \mathbf{X}_{\text{CH9}}] \in \mathbb{R}^{B \times T \times 3 \times H \times W} \quad (5.1)$$

$$\mathbf{X}_{\text{radar-only}} = \mathbf{X}_{\text{radar}} \in \mathbb{R}^{B \times T \times 1 \times H \times W} \quad (5.2)$$

Here, B denotes the batch size, $T = 4$ is the number of input time steps, and H and W represent the spatial dimensions. Thus, the two configurations differed only in their input channel composition, allowing a direct comparison under comparable experimental conditions. Figure 5.4 shows a representative example of the model predictions, while Table 5.2 reports the aggregated quantitative results obtained across the entire test set.

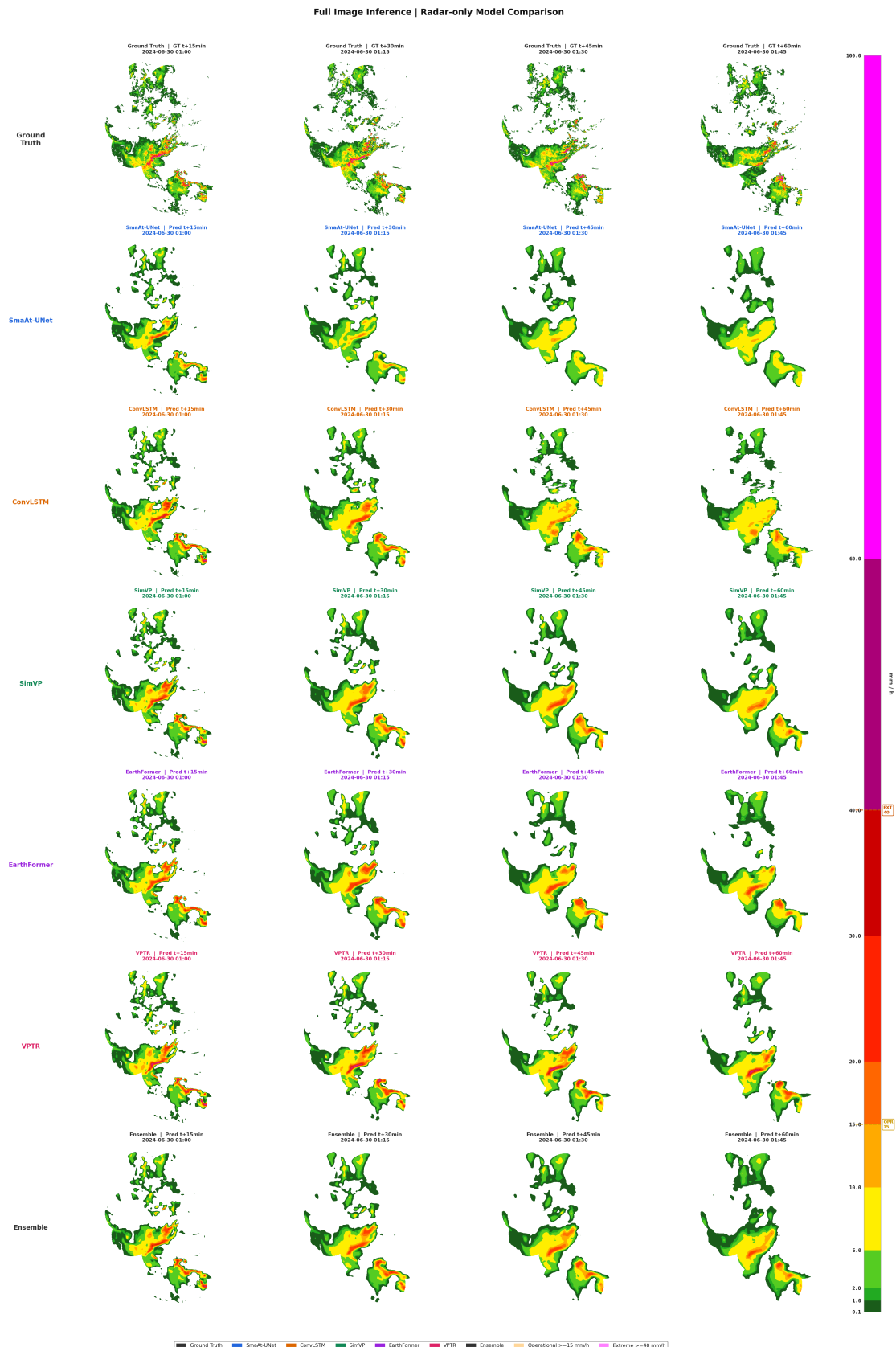


Figure 5.4: Precipitation forecasts from radar-only observations for a representative convective event on 30 June 2024. The ground truth shows several precipitation cells of varying intensity that gradually weaken and shift over the forecast window. Intense cells are generally better captured than lighter, scattered areas, with differences between models becoming more visible at longer lead times. Colour indicates precipitation intensity from light rainfall (■) to extreme convective precipitation (■), with the operational threshold marked on the colour bar.

Table 5.2: Performance comparison across radar-only models and lead times.

Lead Time	Model	ETS $\uparrow$	CSI $\uparrow$	MAE $\downarrow$	MSE $\downarrow$	PSNR $\uparrow$
$t + 15$ min	ConvLSTM	0.20	0.26	0.28	2.90	37.5
	EarthFormer	0.24	0.28	0.25	2.60	38.1
	SimVP	0.26	0.31	0.23	2.30	38.5
	SmaAt-UNet	0.21	0.22	0.38	4.20	35.9
	VPTR	0.29	0.32	0.21	2.20	38.7
	Ensemble	0.25	0.27	0.27	2.84	37.6
$t + 30$ min	ConvLSTM	0.17	0.19	0.34	3.40	36.8
	EarthFormer	0.23	0.25	0.26	2.65	37.9
	SimVP	0.17	0.19	0.26	2.70	37.8
	SmaAt-UNet	0.13	0.14	0.49	<u>5.60</u>	<u>35.1</u>
	VPTR	<u>0.21</u>	<u>0.24</u>	0.29	3.10	37.3
	Ensemble	<u>0.18</u>	<u>0.19</u>	<u>0.32</u>	3.45	36.7
$t + 45$ min	ConvLSTM	0.11	0.12	0.45	5.20	35.3
	EarthFormer	0.14	0.14	<u>0.36</u>	3.90	<u>36.6</u>
	SimVP	<u>0.10</u>	<u>0.15</u>	0.49	<u>5.40</u>	35.2
	SmaAt-UNet	0.05	0.08	0.56	6.80	34.1
	VPTR	0.16	0.18	0.29	3.12	37.2
	Ensemble	0.11	0.13	0.42	4.85	35.6
$t + 60$ min	ConvLSTM	0.10	0.11	0.52	6.10	34.6
	EarthFormer	<u>0.12</u>	<u>0.13</u>	<u>0.41</u>	4.60	<u>35.8</u>
	SimVP	<u>0.07</u>	<u>0.09</u>	<u>0.51</u>	<u>5.56</u>	<u>35.1</u>
	SmaAt-UNet	0.02	0.04	0.64	7.90	33.5
	VPTR	0.13	0.15	0.33	3.50	36.7
	Ensemble	0.10	0.12	0.47	5.48	35.2

Bold indicates the best value for each metric at each lead time, while underlining indicates the second-best value. $\uparrow$ indicates that higher values are better, and $\downarrow$ indicates that lower values are better. Tied values receive the same formatting. All CSI and ETS values are averaged across the 5 and 15mm/h thresholds; the subscript is omitted from the column header due to space constraints.

Similar to the previous section, Figures 5.5 and 5.6 illustrate the radar-only model performance across the four forecast lead times. The categorical results are presented separately from the continuous and image-based measures, making the different evaluation perspectives easier to examine. This presentation also allows the performance patterns of the evaluated architectures to be compared across successive forecast horizons.

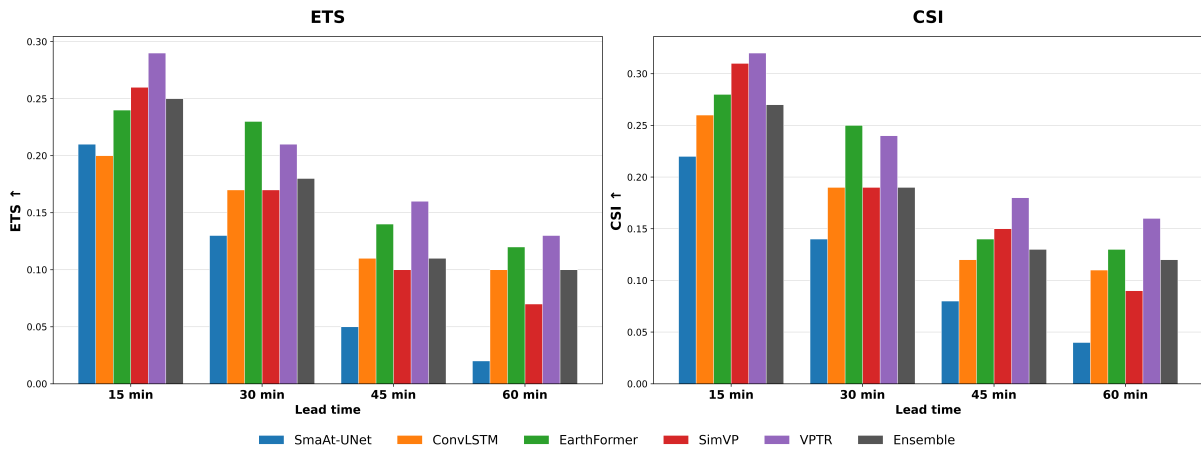


Figure 5.5: Radar-only categorical performance across forecast lead times.

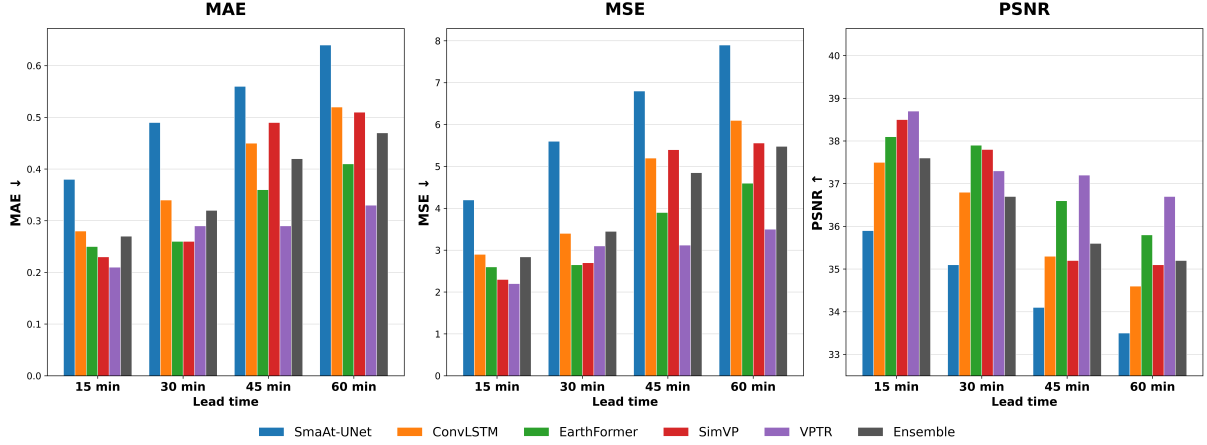


Figure 5.6: Radar-only continuous and image-based performance across lead times.

The quantitative results in Table 5.2 indicate a gradual decline in forecasting performance as forecast lead time increases across all evaluated models. Metrics related to precipitation detection, such as ETS and CSI, decrease with increasing forecast horizon, while reconstruction errors, including MAE and MSE, increase, as shown in Figure 5.6. In addition, a progressive reduction in PSNR is observed at longer forecast lead times, reflecting lower similarity between the predicted and observed precipitation fields. Therefore, this pattern highlights the increasing difficulty of maintaining reliable precipitation forecasts as the prediction horizon extends.

5.3 Non-Deep-Learning Approach

The same input configurations were also evaluated using a non-deep-learning baseline implemented with pySTEPS [3, 12], using the same forecast lead times as the deep learning models. This approach requires no training of learnable parameters; forecasts are produced by estimating a motion field via Lucas-Kanade optical flow and extrapolating the most recent observed radar field forward along it, with satellite data informing only the motion estimate in the multimodal configuration. The extrapolation settings such as motion estimation method, interpolation order, and persistence blending were selected via a small grid search on a held-out validation subset and then applied unchanged to the test subset, summarized in Table 5.3. Its inclusion provides a conventional reference for assessing the benefit of learned spatiotemporal representations.

Table 5.3: Grid search space and selected configuration for the pySTEPS baseline.

Parameter	Search space	Selected
Motion estimation method	LK, VET	LK
Nowcast method	Extrapolation, S-PROG	Extrapolation
Extrapolation interpolation	Nearest, Bilinear	Bilinear
Persistence blending	On, Off	Off

Configuration selected via grid search on a held-out validation subset (disjoint from the test set), scored by pooled CSI and ETS at the 5, and 15 mm/h threshold, averaged over the 15 and 30 minute lead times, then applied unchanged to the test subset.

A qualitative example for a representative convective event is presented in Figure 5.7, shown here as a single illustrative case, and quantitative results aggregated over the complete test set are reported in Table 5.4. Overall, forecast quality declines with increasing lead time: the broad precipitation envelope remains identifiable, but fine-scale structure is smoothed away and small convective cores drift from their true position as lead time increases.

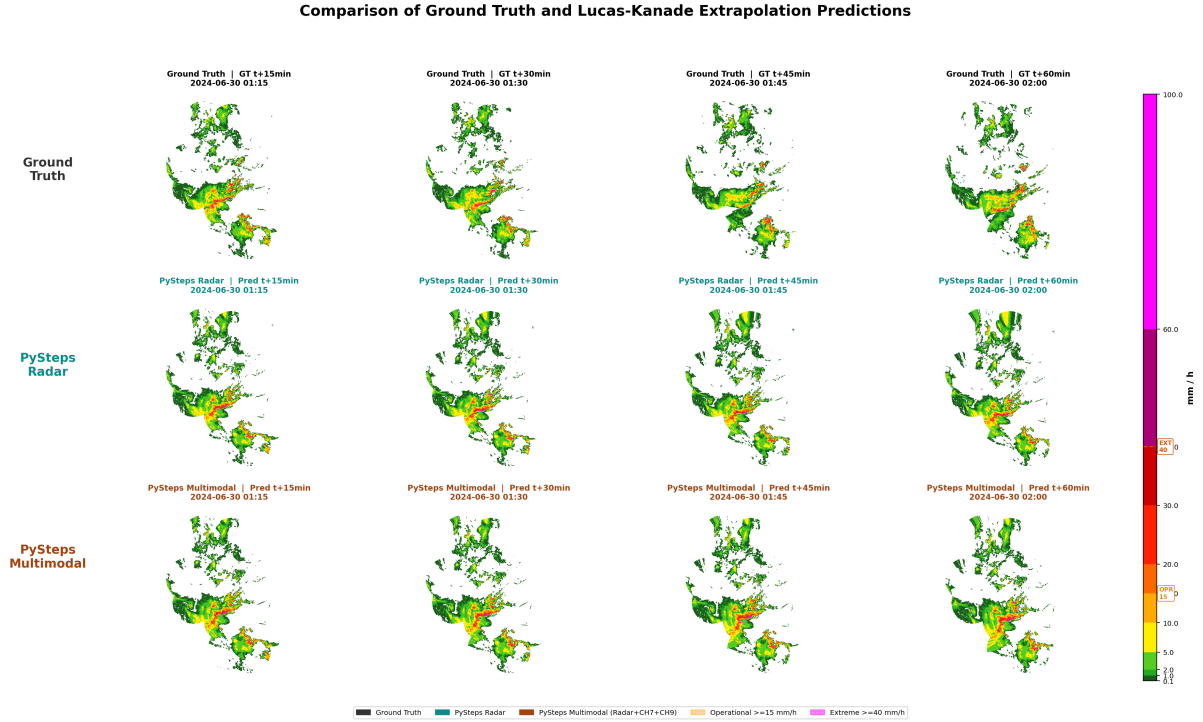


Figure 5.7: Non-deep-learning precipitation nowcasting predictions for the radar-only and multimodal configurations at $t+15$, $t+30$, $t+45$, and $t+60$ forecast horizons for a representative convective event. The colour scale ranges from light rainfall (■) to extreme convective precipitation (■), following the same convention as the previous figures.

Table 5.4: Extrapolation-Based Baseline Performance Across Lead Times.

Lead Time	Model	ETS $\uparrow$	CSI $\uparrow$	MAE $\downarrow$	MSE $\downarrow$	PSNR $\uparrow$
$t+15$ min	Radar	0.28	0.30	0.13	0.94	42.4
	Multimodal	0.21	0.22	0.15	1.21	41.3
$t+30$ min	Radar	0.15	0.17	0.17	1.43	40.6
	Multimodal	0.12	0.12	0.20	1.68	39.9
$t+45$ min	Radar	0.10	0.13	0.20	1.72	39.8
	Multimodal	0.08	0.10	0.23	1.97	39.2
$t+60$ min	Radar	0.08	0.09	0.22	1.84	39.5
	Multimodal	0.06	0.08	0.25	2.11	38.9

Bold indicates the best result for each metric at each lead time. $\uparrow$ indicates that higher values are better, and $\downarrow$ indicates that lower values are better. All CSI and ETS values are averaged across the 5 and 15mm/h thresholds; the subscript is omitted from the column header due to space constraints.

Similar to the previous section, Figures 5.8 and 5.9 illustrate the pySTEPS baseline per-

formance across the four forecast lead times, with categorical and continuous metrics presented separately for clarity.

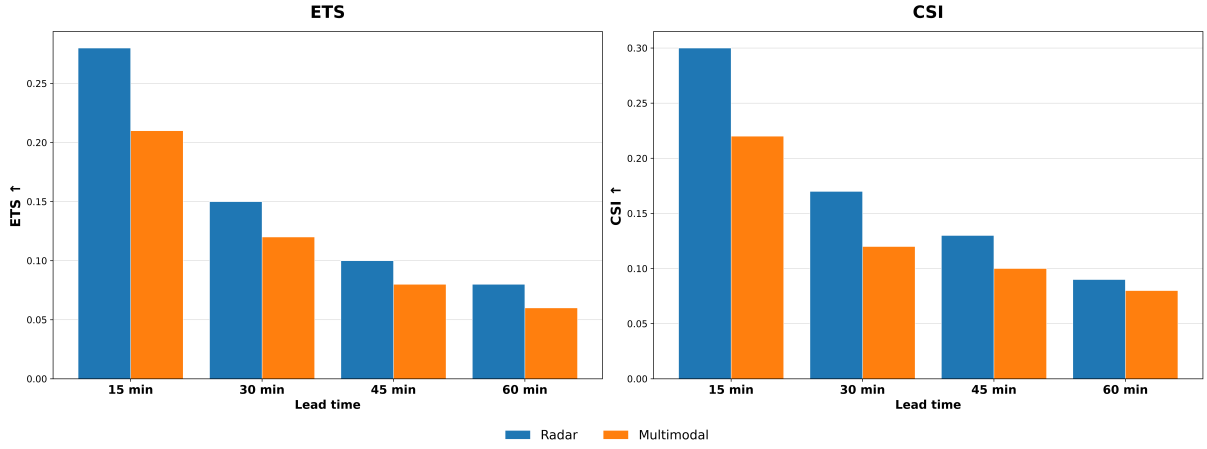


Figure 5.8: Non-DL categorical performance across forecast lead times.

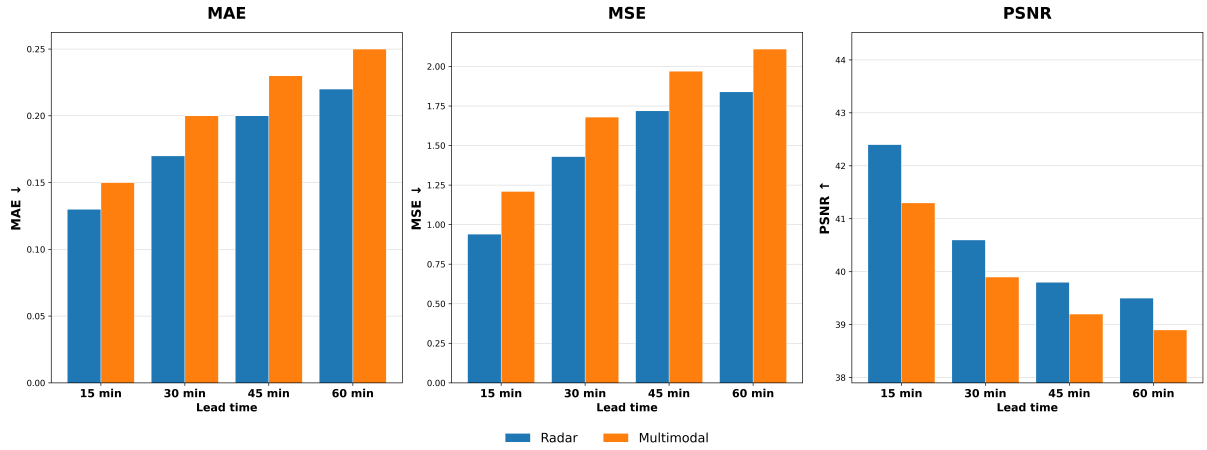


Figure 5.9: Non-DL continuous and image-based performance across lead times.

The results in Table 5.4 show an overall reduction in precipitation detection skill as the forecast horizon increases. ETS and CSI decrease for both configurations across all evaluated lead times. The radar-only configuration performs better across the evaluated forecast horizons, achieving higher ETS and CSI, lower MAE and MSE, and higher PSNR. This indicates that incorporating satellite information does not improve the non-deep-learning baseline forecasts in this experiment. However, PSNR should be interpreted with caution, as it measures pixel-level reconstruction quality rather than the precise detection and location of convective precipitation. Overall, although the radar-only configuration performs better, both configurations show a loss of detection skill with increasing lead time. This decline is consistent with extrapolation-based forecasting limitations (Section 2.2.2), where precipitation changes cannot be captured through motion extrapolation alone.

5.4 Results Overview

The multimodal results in Table 5.1 show that the evaluated architectures exhibit different strengths rather than a single model consistently outperforming the others across all evaluation metrics. VPTR achieves the highest categorical scores, such as ETS and CSI, across most forecast lead times, indicating strong categorical performance over the evaluated horizons. In contrast, the Ensemble shows competitive performance for several reconstruction and image-based metrics at shorter and intermediate forecast horizons, although this performance is not maintained consistently at longer lead times, because averaging cannot correct for errors that the individual models share, which become more pronounced as lead time increases. Overall, the relative performance of the architectures varies depending on both the evaluation metric and the forecast horizon.

Considering the effect of forecast horizon more specifically, forecast accuracy generally decreases with increasing lead time across the evaluated architectures. Although the differences between the models are relatively small at shorter forecast horizons, they become more pronounced beyond 30 minutes. SmaAt-UNet was different in the radar-only configuration, where its performance was much lower than the other architectures at every lead time, because its lightweight, parameter-efficient design (Section 4.2.3) relies more heavily on the additional cloud-top information from the satellite channels to compensate for its reduced capacity. Among the individual models, VPTR demonstrates more consistent performance at longer lead times, indicating better preservation of precipitation detection skill as the forecast horizon increases.

A comparison between the multimodal and radar-only configurations demonstrates the benefit of incorporating satellite observations. As shown in Figures 5.2 and 5.5, the multimodal predictions maintain better spatial coverage and more coherent precipitation structures across all forecast horizons, while the radar-only predictions show more fragmented and spatially inconsistent outputs, particularly at $t+45$ and $t+60$. The pattern observed in this representative event is also reflected in the overall quantitative results reported in Tables 5.1 and 5.2, where the multimodal configuration generally achieves better performance across the evaluated models and forecast lead times. This improvement may be associated with the additional information provided by the CH7 and CH9 satellite channels as changes in cloud-top temperature can indicate developing convection before the associated precipitation becomes clearly detectable by radar. In contrast, a comparison with the extrapolation-based baseline in Figures 5.8 and 5.9 reveals that the extrapolation approach did not benefit from incorporating satellite observations, indicating that the advantage of multimodal inputs depends heavily on the underlying forecasting approach.

Comparing Tables 5.1 and 5.2 with the extrapolation-based baseline in Table 5.4, the pySTEPS radar-only configuration achieved lower MAE and MSE and higher PSNR than the evaluated deep learning models at all forecast lead times, whereas the deep learning models generally achieved higher CSI and ETS. This difference highlights the complementary perspectives provided by continuous and categorical verification metrics in precipitation nowcasting. Continuous pixel-wise metrics such as MAE, MSE, and PSNR are sensitive to the magnitude and spatial distribution of pixel-level errors, whereas categorical metrics such as CSI and ETS evaluate the ability to correctly identify precipitation events at specified thresholds. This distinction is related to the double-penalty problem

in precipitation nowcasting verification [65]. The hybrid loss function used in this study (Section 4.3.1) was designed to place greater emphasis on precipitation intensity and spatial sharpness, particularly for heavy convective events. This design may have contributed to the observed difference between the continuous and categorical evaluation results. Thus, the deep learning models achieved stronger event-detection scores while exhibiting higher pixel-wise errors relative to the extrapolation-based forecasts. This result demonstrates that the choice of evaluation metric can influence the apparent relative performance of different nowcasting approaches.

The differences between the two configurations become more pronounced at longer forecast horizons. However, understanding the specific contribution of the CH7 and CH9 information to the model predictions requires further investigation using XAI techniques, which is considered as part of future work (Section 6.4).

Predicting intense convective precipitation remains the greatest challenge for all evaluated architectures. This is illustrated by the representative event in Figure 5.1, where the models underpredict the most intense convective cores. Although the weighted loss function and balanced sampling strategy (Sections 4.3.1 and 3.5.5) help mitigate class imbalance, they cannot fully compensate for the limited number of high-intensity convective events available during training. Consequently, the most intense convective cells remain the most difficult features to predict accurately, indicating that increasing the representation of such events in the training data may be as important as further architectural improvements.

Chapter 6

Conclusions

This thesis investigated deep learning approaches for short-term convective precipitation nowcasting using radar and satellite observations. Three specific expectations about convective precipitation nowcasting were examined through experiments with different architectures, input configurations, and ensemble approaches. The experiments showed differences in forecasting performance between the tested architectures and input configurations, while the prediction of intense precipitation remained difficult. The three research hypotheses are discussed first, followed by the limitations of the current work, the lessons learned during the project, and possible directions for future research.

6.1 Discussion of Research Hypotheses

H1 proposed that the choice of deep learning architecture affects precipitation nowcasting performance under comparable experimental conditions. The relative ranking of the architectures was not fixed. It shifted depending on the input configuration and the evaluation metric used. At the 60 minute lead time, for instance, SimVP’s ETS score fell from 0.14 with multimodal input to 0.07 with radar-only input (about 50%), while ConvLSTM showed a smaller decline, from 0.12 to 0.10 (about 17%). Across all six models at this lead time, the decline ranged from about 8% (EarthFormer) to about 82% (SmaAt-UNet), suggesting that SimVP and ConvLSTM were representative of this broader spread rather than isolated cases. CSI followed a comparable pattern at the same lead time, with a larger change again observed for SimVP than for ConvLSTM. Architecture choice therefore appears to matter most in the setting where less input information is available, rather than uniformly across all conditions.

H2 proposed that combining radar and satellite observations improves precipitation nowcasting performance compared with using radar observations alone. Across the evaluated deep learning models and lead times, the multimodal configuration generally produced better results than the radar-only configuration, though not without exception. EarthFormer’s ETS score at the 30 minute lead time was one such case, coming in slightly lower under multimodal input, 0.22 against 0.23 (about 4% lower). For the ensemble, the benefit of satellite input widened with lead time: MSE stood at 1.90 against 2.84 at 15 minutes, and by 60 minutes at 2.93 against 5.48. The extrapolation-based baseline broke from this trend, its PSNR fell from 42.4 to 41.3 once satellite input was added

rather than improving. Taken together, these patterns point to an architecture-dependent effect: satellite information appears useful to the deep learning models in ways that the extrapolation-based baseline was not able to exploit.

H3 proposed that an ensemble of multiple deep learning models improves precipitation nowcasting performance compared with individual architectures. The ensemble was constructed by averaging the predictions of all five architectures. It held its own on skill scores, but the advantage over the strongest individual model (VPTR) was inconsistent on error-based metrics. On MAE, the ensemble actually outperformed VPTR at the shorter lead times, by about 6% at $t+15$ and about 10% at $t+30$, but fell behind at longer horizons, trailing by about 5% at $t+45$ and about 12% at $t+60$. On MSE, the ensemble trailed VPTR across all four lead times, by roughly 8% to 15%. At the 60 minute lead time under multimodal input, its MSE reached 2.93 against 2.70 for VPTR (about 9% higher), and MAE reached 0.28 against 0.25 (about 12% higher). The results suggest that simple averaging does not consistently retain the performance of the strongest individual model, particularly for squared-error-based metrics and at longer forecast horizons. Therefore, targeted combination strategies, such as weighted averaging or stacking, may be worth exploring in future work.

6.2 Limitations

The selection of satellite channels limited the amount of atmospheric information available to the models. The experiments were based on two of the 12 spectral channels available from the SEVIRI instrument [37]. The selected channels provide relevant cloud-top information for convective nowcasting, while the remaining channels contain additional atmospheric information that was not used. For example, the water-vapour channels provide information about moisture at different tropospheric levels, which is closely related to convective initiation and storm development. The additional channels were excluded because their collection and preprocessing would have required more time and computational resources.

The available inputs were further limited to radar and satellite observations. Additional meteorological variables, such as temperature, humidity, wind speed and direction, solar radiation, and lightning data, have been used in convective precipitation nowcasting [66]. These additional sources were not included because developing a reliable multi-source fusion pipeline requires substantial time and careful engineering. Consequently, the models did not have access to several atmospheric variables that could provide information about conditions associated with convective development.

The evaluation was limited to the 2024 test period (Table 3.10), so it does not capture how convective precipitation varies between different years. Weather conditions and precipitation patterns can change from year to year, which may affect model performance under conditions that are not well represented in the test data. The training, validation, and test datasets were limited to the months of June to September because convective precipitation occurs more frequently and is usually more intense during this period. As a result, the models were not evaluated under conditions from the other months of the year, and their performance across different seasons remains uncertain. Evaluating the models over several years and including different seasons would provide a more complete

assessment of their ability to generalize across different years and seasonal conditions.

The range of loss function formulations considered in this study was limited. Alternative formulations, such as dice loss, focal loss, or quantile-based losses for probabilistic prediction, were not evaluated. These formulations were excluded because comparing multiple loss functions across five architectures would have required additional training time and computational resources. The evaluation was also limited to selected metrics, so other aspects of forecast performance were not considered. As a result, it remains unclear whether alternative loss formulations could have further improved the predictive performance of the models.

The available computational resources limited the range of models and experiments included in the comparison. Training was limited to a maximum of 50 epochs with early stopping, which may have limited further optimisation of models that could benefit from longer training. Five deterministic deep learning architectures were evaluated, while generative adversarial networks [20] and diffusion-based models [22] were not considered because of their computational and training requirements. Transfer learning approaches were also not investigated, as the models were trained from scratch rather than initialized from existing pretrained weights [67]. For the extrapolation-based baseline, Lucas-Kanade optical flow was used as the motion estimation method, while other techniques, such as variational methods (VET) and the S-PROG cascade model, were not extensively compared. As a result, the comparison does not cover the full range of generative, transfer learning, and traditional extrapolation-based approaches.

The comparisons between the models were based only on the reported performance metrics, and no statistical significance testing was carried out. As a result, small differences between the models should not be read as statistically significant improvements. Future work could apply statistical tests or confidence intervals to give a clearer picture of the uncertainty in the reported results.

6.3 Lessons Learned

Working on this project showed that decisions made at an early stage do not always remain suitable throughout the research process. Some initial choices had to be reconsidered after the first experiments, when their limitations became clearer. Being flexible with these decisions made it possible to adjust the approach based on the results and observations from the experiments.

Working with a supervisor experienced in atmospheric science was also valuable. Several modelling decisions and the interpretation of unexpected results benefited from meteorological knowledge beyond a purely machine learning perspective. This experience showed how domain knowledge can support technical decisions and help produce results that are more meaningful from a meteorological perspective.

The experimental process also involved several attempts that did not produce the expected improvements. However, these experiments were still useful because they revealed problems that were difficult to identify beforehand. Understanding why an approach did not work helped guide the changes made in later experiments and provided a better understanding of the modelling process.

Time management became particularly important as the project progressed. Training the models was only one part of the workload, while debugging, repeating experiments, analysing results, and documenting the work also required considerable time. For a similar project, more time should therefore be reserved for these activities and for unexpected problems that can arise during the research process.

6.4 Future Work

Additional SEVIRI spectral channels could be included to capture more information about atmospheric conditions during storm development. Lightning observations and numerical weather prediction fields could also be added as model inputs. These variables could provide information about atmospheric conditions that are not available from the current radar and satellite inputs.

The current study combined radar and satellite observations at the input stage. Future work could investigate alternative fusion strategies in which each data source is processed separately before the extracted features are combined. Feature-level or intermediate fusion methods could also be explored to determine whether a different integration approach improves forecasting performance. Keeping the other experimental conditions unchanged would help isolate the effect of the fusion strategy.

The prediction of intense convective precipitation remains an area for improvement. Alternative sampling methods could be tested to increase the representation of intense events during training. The dataset could also be extended with additional convective seasons or data from other regions to provide more examples of these less frequent precipitation patterns.

The ensemble predictions in this study were obtained by averaging the outputs of the individual models. Future work could investigate weighted ensemble strategies in which different models receive different weights based on their forecasting performance. The weights could also be determined separately for different forecast horizons to account for changes in model performance with lead time.

The models used in this thesis operate as black boxes and provide limited insight into how individual inputs influence their predictions. Future work could apply XAI techniques, such as saliency maps or attention visualisation, to identify the features and spatial patterns that contribute most to a forecast. This could help determine how the models use satellite information before convective development becomes visible in radar observations.

Evaluating the models over multiple years and individual seasons would show whether their performance changes under different weather conditions. Since the dataset covers a single geographic region, testing the models in regions with different rainfall characteristics could also show how well the results transfer to other areas. Spatial verification metrics such as the Fractions Skill Score could additionally be used alongside the Probability of Detection (POD) and False Alarm Ratio (FAR) to assess forecast performance at different spatial scales and for different storm intensities.

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